



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

**EduScribe: An AI-Assisted Web-Based Lesson Planning
and Academic Resource Management System for
Dela Paz National High School**

A Capstone Project
Presented to the Faculty of the College of Information Technology and Engineering
Polytechnic University of the Philippines-Biñan Campus
Biñan City, Laguna

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Information Technology

By:
Alicaway, Ryza Mae L.
Aquino, Dan Uziel B.
Hermosa, Philip Ceasar G.
Regala, Mary Grace A.

May 2026



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

CHAPTER 1

THE PROBLEM AND ITS SETTING

Introduction

Education plays a significant role in the development of individuals and society by providing knowledge, skills, values, and competencies necessary for personal growth and national progress (Crompton & Burke, 2023). Across the world, educational institutions continuously seek effective strategies and innovative approaches to improve instructional delivery, student learning outcomes, and academic management practices. As societies become increasingly dependent on digital technologies, the education sector has also experienced rapid transformation through the integration of technological tools and systems that support teaching, learning, communication, and administrative operations. These developments have encouraged educational institutions to adopt modern solutions that improve efficiency, accessibility, and the overall quality of education. The growing influence of technology in education has become particularly evident in countries that aim to modernize their educational systems and address the changing needs of learners and educators in the digital age (Teräs et al., 2022; UNESCO, 2023).

In the Philippines, the educational sector continuously adapts to curriculum reforms, technological advancements, and evolving instructional demands in order to improve the quality of education and promote more effective learning environments (Barrot, 2021). Educational institutions are expected to provide



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

efficient academic systems that support both teachers and students while maintaining compliance with institutional and national educational standards. Teachers, in particular, perform multiple responsibilities that extend beyond classroom instruction, including lesson planning, preparation of teaching materials, assessment development, student performance monitoring, documentation, and coordination of academic requirements. These responsibilities require substantial time, effort, organization, and consistency to ensure that instructional objectives are effectively achieved. As educational demands continue to increase, many educators experience difficulties in balancing instructional responsibilities with administrative workloads, resulting in stress, inefficiency, and reduced productivity. This growing complexity within the educational environment highlights the importance of integrating technological solutions that can assist educators in managing academic tasks more effectively.

The integration of technology in the education sector has become one of the primary approaches in addressing instructional and administrative challenges encountered by schools and educators (Haleem et al., 2022; UNESCO, 2023). Technological innovations have enabled educational institutions to improve communication, automate processes, organize academic resources, and provide more accessible learning environments for students and teachers. Through the use of digital platforms, schools are able to streamline educational operations and reduce dependence on manual processes that often consume significant time and effort. Technology also supports collaboration, accessibility, and flexibility by



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

allowing users to access educational materials and academic information through internet-connected devices such as computers, laptops, tablets, and mobile phones. As a result, educational institutions continue to explore and implement technology-based systems that can improve both instructional efficiency and academic management practices.

One of the most widely adopted technological solutions in educational institutions is the use of web-based systems for academic management and instructional support. Web-based educational systems provide centralized platforms where users can access academic information, organize learning materials, manage assessments, and facilitate communication within the school environment. These systems are designed to improve accessibility and efficiency by allowing users to interact with academic resources through online platforms rather than relying solely on traditional paper-based methods. Unlike conventional systems that often operate independently or require manual processing, web-based systems promote real-time access, centralized organization, and collaborative academic workflows. Through these capabilities, educational institutions are able to improve operational efficiency and support more organized academic processes (Crompton & Burke, 2023; Haleem et al., 2022).

Modern web-based educational systems also provide advanced functionalities that support resource management, data organization, and instructional coordination. Features such as centralized repositories, cloud-based storage, collaborative editing, automated organization, and performance monitoring



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

allow educators to manage academic resources more effectively and efficiently. These systems reduce fragmentation by consolidating various academic functions into a unified platform that supports lesson planning, communication, assessment management, and resource accessibility. Furthermore, web-based systems improve consistency and coordination among educators by providing structured environments where instructional materials and academic documents can be systematically organized and retrieved. The increasing use of these systems demonstrates the growing importance of digital technologies in supporting academic management and instructional delivery within educational institutions (Haleem et al., 2022).

Despite the continuous advancement of educational technologies, many schools still experience challenges in managing lesson planning and academic resources due to reliance on traditional and manual methods. Lesson planning remains one of the most important yet time-consuming responsibilities of educators because it requires the preparation of instructional objectives, learning activities, assessment strategies, and teaching materials aligned with curriculum standards. Teachers are expected to ensure that lesson plans are comprehensive, organized, and responsive to the learning needs of students while also maintaining institutional documentation requirements. In addition to lesson planning, educators are also responsible for organizing worksheets, monitoring student performance, preparing instructional materials, and managing classroom records. These responsibilities



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

create substantial workloads that require efficient organization and effective time management.

Traditional lesson planning approaches are often characterized by repetitive documentation, manual encoding of instructional content, and fragmented organization of academic resources. In many educational institutions, teachers still rely on paper-based methods or disconnected software applications that lack integration and centralized functionality. As a result, educators are required to manage multiple tools and platforms simultaneously, leading to inefficient workflows, duplicated efforts, and inconsistencies in academic documentation. These traditional methods consume significant time that could otherwise be allocated to instructional improvement, classroom interaction, and student engagement. Furthermore, manual systems increase the possibility of data loss, disorganization, and delays in accessing important instructional materials and academic records (Belloula, 2025).

The challenges associated with traditional lesson planning and academic management become more evident as teachers attempt to balance instructional quality with administrative responsibilities. Educators often experience difficulty maintaining organized learning materials, tracking instructional progress, and ensuring curriculum alignment due to fragmented systems and limited technological support. This situation is particularly challenging for novice teachers who are still developing pedagogical competence while adapting to institutional requirements and teaching demands. The absence of integrated systems capable of supporting



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

academic organization and instructional preparation contributes to increased workload, reduced efficiency, and inconsistent academic management practices. Consequently, there is a growing need for centralized systems that can streamline instructional workflows and improve the organization of academic resources within educational institutions.

To address these limitations, educational institutions have gradually adopted web-based academic management systems that support lesson planning, resource organization, communication, and instructional coordination within a centralized environment. These systems provide educators with structured digital platforms that simplify academic workflows and improve accessibility to instructional resources. Through centralized management features, teachers can efficiently organize lesson plans, store instructional materials, manage assessments, and monitor academic activities using a single platform. Web-based systems also improve collaboration among users by enabling real-time access to information and academic documents. These capabilities significantly reduce inefficiencies associated with fragmented systems and manual documentation processes while supporting more organized instructional practices (Haleem et al., 2022).

The evolution of educational technologies has further introduced Artificial Intelligence (AI) into academic management systems to enhance instructional support and improve operational efficiency. Artificial Intelligence refers to technologies capable of performing intelligent tasks such as analyzing data, generating content suggestions, automating repetitive processes, and assisting



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

users in decision-making activities. In educational settings, AI-assisted systems are increasingly utilized to support lesson planning, resource management, assessment generation, and instructional organization. These technologies are designed to function as assistive tools that enhance educator productivity while maintaining teacher control over instructional decisions and pedagogical approaches. The integration of AI into educational systems reflects the continuing effort to modernize instructional practices and reduce administrative burdens within schools (Crompton & Burke, 2023).

AI-assisted lesson planning systems are capable of generating instructional recommendations such as lesson objectives, learning activities, worksheets, assessment suggestions, and curriculum-aligned content based on user input and educational standards (Kasneci et al., 2023). These intelligent systems help educators reduce the time required for repetitive planning tasks while improving the consistency and organization of instructional materials. Through AI integration, teachers are provided with additional support in preparing lesson content and managing academic resources more efficiently. Rather than replacing teachers, AI technologies enhance instructional workflows by assisting educators in performing routine academic tasks and allowing them to focus more on instructional delivery, classroom interaction, and student learning outcomes (Bora & Kölemen, 2025).

In addition to supporting lesson planning, AI-assisted web-based systems also improve academic resource management through automated organization, centralized storage, and intelligent data processing capabilities. These systems



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

allow educators to efficiently categorize, retrieve, and manage instructional materials within accessible digital environments. Features such as automated formatting, resource recommendations, progress monitoring, and data organization contribute to improved efficiency and reduced administrative workload. AI-supported platforms also provide scalability and flexibility that accommodate the growing academic and technological needs of educational institutions. Through these advancements, educators are able to maintain more organized academic workflows while improving accessibility and coordination of instructional resources (Kovari, 2025).

Several studies have emphasized the effectiveness of AI-assisted educational systems in improving instructional efficiency and reducing teacher workload. Research involving educators demonstrated that AI-integrated systems significantly reduce lesson preparation time, minimize administrative burden, and improve instructional planning efficiency compared to traditional methods (Bora & Kölemen, 2025). Similarly, studies on intelligent educational platforms revealed positive impacts on academic organization, teaching effectiveness, and resource management within learning environments (Kovari, 2025). Other international studies also highlighted that web-based lesson planning systems contribute to improved productivity, accessibility, and instructional coordination among educators (Belloula, 2025). These findings support the growing adoption of AI-assisted technologies in education and demonstrate their potential to enhance instructional management practices.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Despite these technological developments, many educational institutions still encounter difficulties related to fragmented systems, inefficient workflows, manual processes, and limited integration of academic management functions. Existing systems often lack comprehensive features capable of supporting lesson planning, instructional material organization, assessment management, and centralized academic coordination within a single platform. In many cases, educators are still required to utilize separate applications and manual methods that increase workload and reduce overall efficiency. These limitations continue to affect instructional preparation, organization of academic resources, and accessibility of learning materials within school environments. Consequently, there remains a need for more integrated, intelligent, and user-centered academic management systems that can effectively support educators in performing instructional and administrative responsibilities.

In response to these identified challenges, this study proposes the development of the EduScribe Portal: An AI-Assisted Web-Based Lesson Planning and Academic Management System intended for Dela Paz National High School. The proposed system is designed to assist teachers in generating lesson plans, organizing instructional materials, managing academic resources, and improving instructional workflows through a centralized web-based platform integrated with Artificial Intelligence technologies. The system aims to provide AI-assisted features capable of generating lesson objectives, instructional activities, worksheets, and assessment suggestions while maintaining educator control over instructional



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

decisions and pedagogical practices. Through centralized academic management and intelligent system functionalities, the EduScribe Portal seeks to improve efficiency, organization, accessibility, and instructional quality within the academic environment.

Theoretical Framework

This study is anchored on the Unified Theory of Acceptance and Use of Technology (UTAUT), which explains how users accept and utilize technology based on factors such as performance expectancy, effort expectancy, social influence, facilitating conditions, and behavioral intention. According to the theory, users are more likely to adopt and use a technological system when they perceive it as useful, easy to use, supported by their environment, and beneficial to their performance. In educational settings, technology acceptance plays an important role in determining the effectiveness and sustainability of digital learning tools and academic management systems.

The EduScribe system is expected to support teachers and students in improving lesson planning, academic organization, and access to instructional resources through an AI-assisted web-based environment. Through the integration of intelligent features and centralized academic management tools, the system aims to enhance user acceptance and improve instructional efficiency within the school environment.

The study is also supported by Digital Learning Theory, which emphasizes the importance of integrating technology in education to enhance instructional



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

delivery, accessibility, collaboration, and learner engagement. Digital learning environments allow teachers and students to access educational materials, communicate, and accomplish academic tasks through internet-based platforms and digital tools. This theory supports the use of web-based systems in education as tools that promote flexible, accessible, and learner-centered learning experiences.

In the context of this study, the EduScribe system provides centralized access to lesson plans, worksheets, announcements, and academic resources that support organized and efficient instructional processes. Through digital learning technologies, the system promotes improved accessibility and instructional support for both teachers and students.

In addition, this study is guided by Systems Theory in Information Systems, which explains that an organization functions effectively when all its components, processes, and users operate as an integrated and interconnected system. Systems Theory emphasizes coordination, interaction, and interdependence among system components to achieve efficiency and consistency. In educational institutions, fragmented systems often lead to inefficiencies, duplication of work, and disorganized academic processes.

The EduScribe system addresses these issues by integrating lesson planning, academic resource management, AI-assisted tools, announcements, and academic monitoring into a single centralized platform. Through system integration, the platform

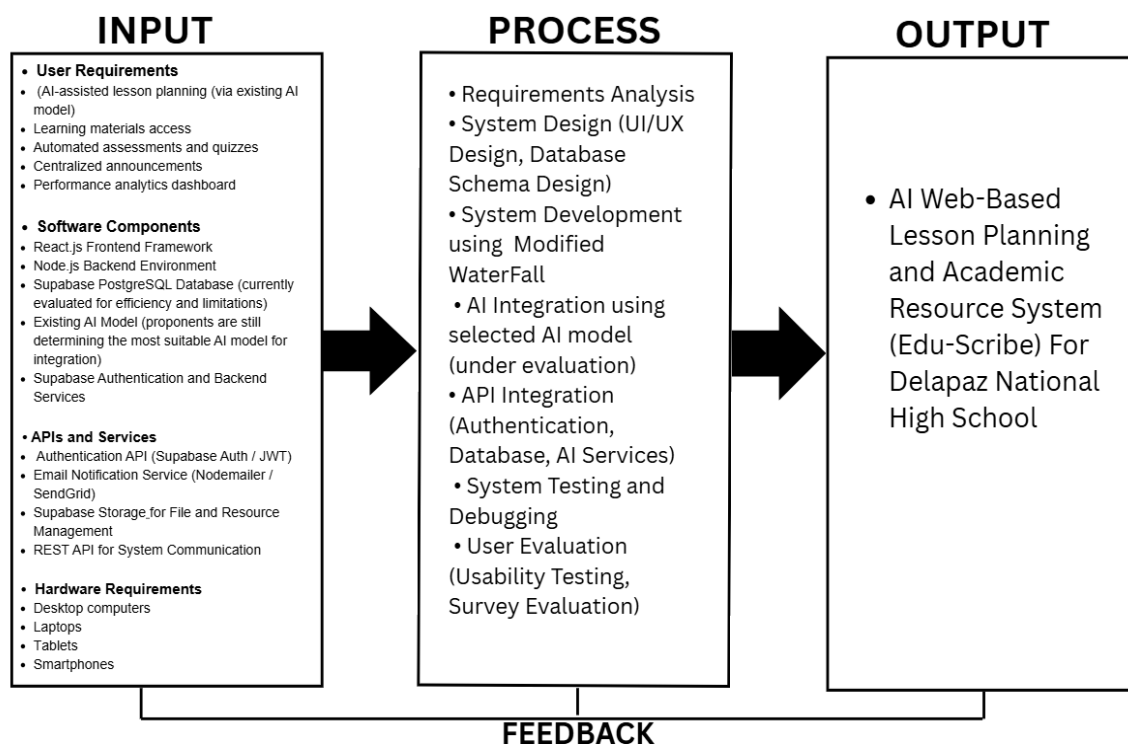


POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

is expected to improve coordination, efficiency, accessibility, and consistency in academic management within Dela Paz National High School.

The integration of these theories provides the foundation for the development of the EduScribe system. UTAUT supports the study in understanding technology acceptance and user behavior, Digital Learning Theory supports the implementation of technology-enhanced educational environments, while Systems Theory supports the integration and coordination of academic processes. Collectively, these theories establish the conceptual and technological foundation for developing an AI-assisted web-based lesson planning and academic management system that addresses the instructional and organizational needs of teachers and students.

Conceptual Framework





POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Figure 1.1

Conceptual Framework showing input, process, and output of the EduScribe Portal

Figure 1.1 illustrates the conceptual framework of the study, which follows the Input–Process–Output (IPO) model. This framework presents the logical and systematic flow of the research in developing the EduScribe Portal: An AI-Assisted Web-Based Lesson Planning and Academic Management System. The IPO model serves as a guide in identifying the necessary inputs, the processes involved in system development, and the expected outputs of the study. It also demonstrates how the identified variables interact to produce an effective and functional academic management solution through the EduScribe Portal.

The Input component of the framework consists of user requirements, software requirements, and hardware requirements. User requirements include the needs of teachers, students, and administrators such as AI-assisted lesson planning, learning material accessibility, centralized announcements, assessments, and performance monitoring. Software requirements involve the development tools, frameworks, artificial intelligence technologies, database systems, and APIs necessary for developing a responsive, secure, and scalable web-based platform. Hardware requirements include computers, laptops, mobile devices, tablets, and servers that support system accessibility and real-time functionality.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

The Process component describes the systematic procedures involved in designing, developing, implementing, and evaluating the EduScribe Portal using the Modified Waterfall Model. The process begins with requirements analysis, where the researchers identify the challenges encountered in lesson planning and academic resource management as well as the system specifications necessary for development. This is followed by system design, which includes the development of interface layouts, workflows, database structures, and overall system architecture to ensure functionality and usability.

The next stage is system development, where the researchers develop the AI-assisted and academic management functionalities of the platform. The implementation phase integrates lesson planning tools, centralized resource management, AI-assisted content generation, announcements, and user authentication features into the system. After implementation, testing and evaluation are conducted to assess the performance, reliability, security, and usability of the system. The evaluation process utilizes selected ISO/IEC 25010 software quality characteristics including usability, functional suitability, performance efficiency, reliability, and security to determine the overall effectiveness and acceptability of the EduScribe Portal.

The Output component represents the expected result of the study, which is the fully developed EduScribe Portal: An AI-Assisted Web-Based Lesson Planning and Academic Management System for Dela Paz National High School. The proposed system is expected to improve lesson preparation, organization of learning materials,



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

accessibility of academic resources, and instructional workflow efficiency for teachers and students. Through AI-assisted functionalities and centralized academic management features, the system aims to reduce administrative workload, improve instructional organization, and enhance the overall academic management process within the institution.

Statement of the Problem

This study aims to develop an EduScribe Portal: An AI-Assisted Web-Based Lesson Planning and Academic Management System for Dela Paz National High School.

Specifically, this study seeks to answer the following questions:

1. What challenges are encountered by teachers in traditional lesson planning and academic resource management in terms of:
 - 1.1 lesson preparation;
 - 1.2 organization of learning materials;
 - 1.3 accessibility of academic resources; and
 - 1.4 management of instructional documents?
2. How can a user-friendly web-based platform be designed to integrate lesson planning, learning materials, assessments, and announcements into a centralized system?
3. How can AI-assisted features and centralized academic resource management functionalities be developed and implemented to support lesson



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

preparation, organization, accessibility, and management of academic resources?

4. How acceptable is the EduScribe Portal based on the selected ISO/IEC 25010 software quality characteristics in terms of:

4.1 functional suitability;

4.2 usability;

4.3 performance efficiency;

4.4 reliability; and

4.5 security?

Statement of Objectives

The objective of this study is to design, develop, implement, and evaluate an EduScribe Portal: An AI-Assisted Web-Based Lesson Planning and Academic Management System intended to improve lesson preparation, academic organization, and accessibility of learning materials for teachers and students of Dela Paz National High School.

Specifically, this study aims to:

- Design a user-friendly web-based platform that integrates lesson planning, learning materials, assessments, and announcements into a centralized system.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Develop AI-assisted features capable of generating lesson objectives, instructional activities, worksheets, and assessment suggestions to support teachers in lesson preparation and academic resource management.
- Implement a centralized academic resource management system with role-based access mechanisms and Learner Reference Number (LRN) authentication to ensure secure storage, organization, accessibility, and management of learning materials and academic files.
- Evaluate the EduScribe Portal using selected ISO/IEC 25010 software quality characteristics, particularly functional suitability, usability, performance efficiency, reliability, and security, based on feedback from intended users.

Scope and Limitations of the Study

This study focuses on the development and implementation of the EduScribe Portal: An AI-Assisted Web-Based Lesson Planning and Academic Management System intended for teachers, students, and administrators of Dela Paz National High School. The system is designed to improve lesson preparation, organization of learning materials, and academic communication through features such as AI-assisted lesson generation, worksheet creation, centralized storage of academic resources, student progress monitoring, and announcement management.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

The system primarily emphasizes AI-assisted lesson planning and worksheet generation as its core functionalities. Additional features such as announcements, monitoring, and resource accessibility are limited to essential implementation to maintain the feasibility of the project within the two-semester capstone development period. The platform can be accessed through web browsers using computers, laptops, tablets, and mobile phones with internet connectivity, allowing users to conveniently utilize the system in different learning environments.

Furthermore, the study includes the evaluation of the system using selected ISO/IEC 25010 software quality characteristics, specifically usability, functional suitability, performance efficiency, reliability, and security, to determine the effectiveness and quality of the developed platform.

However, the study has several limitations. The system is limited to core academic management functions and does not include advanced functionalities such as a full-scale Learning Management System (LMS), comprehensive grading system, attendance management, or student information system. The performance of the system is dependent on stable internet connectivity, and some features may not function efficiently under poor network conditions.

Additionally, the AI-assisted functionalities are limited to generating suggestions and recommendations for lesson preparation and do not fully replace the professional judgment and decision-making of teachers. System performance and user experience may also be affected by older devices, limited hardware



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

specifications, or slow internet connections. The study is further limited to the intended users within Dela Paz National High School and may require additional modifications before implementation in other educational institutions.

Significance of the Study

The project focuses on developing an AI-assisted web-based lesson planning and academic management system. This study is expected to benefit the following:

Teachers. The system will help teachers reduce time and effort in preparing lesson plans, organizing instructional materials, and creating assessments. Through AI-assisted tools and centralized academic management features, teachers can improve instructional efficiency, maintain organized learning resources, and focus more on classroom engagement and student learning outcomes.

Students. Students will benefit from improved accessibility to learning materials, worksheets, announcements, and academic requirements through a centralized platform. The system will support independent and self-paced learning by providing organized and accessible educational resources that students can access anytime using internet-connected devices.

School Administrators. The system will provide administrators with a centralized academic management platform that supports monitoring of academic activities, organization of instructional resources, and coordination of school-related



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

processes. This can improve academic supervision, decision-making, and overall management efficiency within the institution.

School Community. The EduScribe Portal will contribute to a more organized, efficient, and technology-supported educational environment by improving communication, coordination, and accessibility among teachers, students, and administrators. The implementation of the system is expected to promote better academic workflows and support the continuous integration of technology in education.

Future Researchers. This study may serve as a reference for future researchers who intend to develop or improve AI-assisted educational systems and web-based academic management platforms. The findings of the study may contribute to future innovations and research related to educational technology, lesson planning systems, and Artificial Intelligence integration in academic environments.

Definition of Terms

Artificial Intelligence (AI)- Refers to the technology that enables the system to generate lesson objectives, instructional activities, assessment suggestions, and other academic recommendations automatically. AI assists teachers in improving lesson preparation efficiency and reducing repetitive manual tasks.

Academic Management- Refers to the process of organizing, monitoring, and managing academic activities such as lesson planning, assessments,



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

instructional materials, and student performance tracking within an educational institution.

AI-Assisted Lesson Generation- Refers to the capability of the system to utilize Artificial Intelligence technologies in generating lesson objectives, learning activities, worksheets, and assessment suggestions based on curriculum requirements and user input.

Analytics- Refers to the process of collecting, analyzing, and interpreting academic data within the system to support monitoring of student performance and instructional effectiveness.

Centralized Repository- Refers to a storage feature within the system where lesson plans, instructional materials, worksheets, and academic files are systematically organized and accessed through a single platform.

Digital Learning Materials- Refers to educational resources stored and accessed in digital format, including lesson plans, worksheets, instructional content, and assessment materials available within the system.

Learner Reference Number (LRN)- Refers to the unique and permanent identification number assigned to students by the Department of Education, which is utilized in the system for secure student authentication and access to learning materials.

Learning Management System (LMS)- Refers to a digital platform designed to manage, organize, deliver, and monitor educational content, assessments, and



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

academic activities within a learning environment.

Lesson Planning- Refers to the process of organizing instructional objectives, learning activities, teaching materials, and assessments to support effective teaching and learning.

Most Essential Learning Competencies (MELC)- Refers to the standardized learning competencies prescribed by the Department of Education that guide teachers in developing curriculum-aligned instructional content and assessments.

System Integration- Refers to the process of connecting various technological components, databases, Artificial Intelligence functionalities, and interfaces into a unified and efficient academic management platform.

User Interface (UI)- Refers to the visual layout, design, and interactive components of the system that enable users to navigate and utilize the platform effectively.

Web-Based System- Refers to a system that operates through the internet and can be accessed using web browsers on devices such as computers, laptops, tablets, and mobile phones.

User Authentication- Refers to the process of verifying a user's identity before granting access to the system to ensure security and authorized use only. It typically involves login credentials such as username and password.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

CHAPTER 2

REVIEW OF LITERATURE AND STUDIES

Foreign Literature

Artificial Intelligence in Education

The integration of artificial intelligence in educational administration has consistently demonstrated significant improvements in efficiency, accuracy, and workload reduction among educators. Nelly Rahmawati Ningsih (2026) found that AI applications in grading, attendance monitoring, and academic documentation management substantially reduce teachers' administrative burden while improving accuracy and transparency in school records.

Similarly, Shanti Ramalingam and Mahendran Maniam (2024) identified key AI-driven trends such as automated administrative processes, predictive analytics, and personalized learning systems that collectively streamline institutional operations. These findings confirm that AI is increasingly being adopted not only as a supplementary tool but as a core component of educational administration systems that handle multiple institutional functions simultaneously.

In addition to administrative efficiency, artificial intelligence has also shown strong contributions in improving instructional design, lesson planning, and educational content generation. Frank Kehoe (2023) explored the use of generative AI tools such as ChatGPT and Google Bard in initial teacher education and found



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

that these systems can produce structured lesson plans and tailored instructional materials based on specific learning objectives, significantly reducing preparation time while maintaining instructional quality.

Supporting this, Rommel Alali (2024) found that generative artificial intelligence has strong potential in automating key academic tasks such as lesson planning, feedback generation, and personalized learning pathways. The study explained that AI can generate structured instructional materials that significantly reduce teachers' preparation time while improving consistency in content delivery. However, it also emphasized that the effectiveness of these tools depends on ethical implementation to ensure accuracy, appropriateness, and alignment with educational standards.

Furthermore, Mujiono Mujiono (2023) emphasized that the effectiveness of artificial intelligence in education is maximized when it works in collaboration with teachers rather than replacing them. The study highlighted that combining AI efficiency with teacher judgment improves learning outcomes by allowing educators to focus more on instructional strategies and student engagement while AI handles repetitive tasks. It also stressed that teachers play a critical role in interpreting and refining AI-generated outputs to ensure instructional relevance.

However, Wai Yie Leong (2025) and Rania Lampou (2023) both pointed out that despite the benefits of AI in education, its implementation comes with challenges such as ethical concerns, data privacy issues, and the risk of over-reliance on automated systems. They also emphasized the importance of proper



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

training and awareness among educators to ensure responsible use of AI technologies. Without these safeguards, AI systems may lead to misuse or reduced critical thinking in instructional processes. These findings highlight the need for balanced implementation strategies that prioritize both efficiency and responsibility.

Machine Learning in Education

The application of machine learning in educational settings has consistently demonstrated strong potential in analyzing student data and supporting data-driven decision-making in academic environments. Zara Ersozlu (2024) found that most research utilizes supervised machine learning techniques such as decision trees, support vector machines, random forests, and logistic regression to predict student performance and identify learning patterns.

Similarly, Juntao Chen (2025) reported that machine learning applications significantly improve student performance prediction, engagement analysis, and retention monitoring from 2016 to 2024. These findings confirm that machine learning is increasingly used not only for prediction but also for generating insights from complex educational datasets. Building on these predictive capabilities, studies have identified specific machine learning algorithms that are commonly used and highly effective in educational contexts.

In addition to algorithm selection, research highlights the importance of feature engineering and data preprocessing in improving the effectiveness of



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

machine learning models in education. Sebastianus Radhya (2022) found that improving prediction accuracy depends on using relevant predictors and applying techniques such as feature selection and data mining. Similarly, S. Oppong (2023) reported that a majority of studies utilize feature selection methods to enhance model performance and accuracy. Dr. P. C. Khanzode (2026) further emphasized that analyzing academic, behavioral, socio-economic, and psychological factors plays a critical role in improving prediction quality.

However, despite its advantages, the application of machine learning in education also presents several challenges that must be addressed to ensure effective and responsible implementation. Dr. Girish Chhimwal and Dr. Sandhya Sinha (2025) identified key issues such as data quality, feature selection limitations, and ethical concerns in educational data mining. Similarly, Dr. P. C. Khanzode (2026) highlighted gaps including reliance on synthetic data, lack of real-world validation, and privacy risks, while recommending the use of Explainable AI and privacy-preserving techniques.

Online Lesson Planning

The integration of digital tools in online lesson planning has demonstrated significant potential in improving teacher efficiency and enhancing the overall quality of instructional plan development. Windi Azahra and M. Muliana (2025) found that technology integration in lesson planning enables teachers to design more structured, flexible, and student-centered lesson plans that align with 21st-century



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

learning requirements, particularly through improved collaboration, personalization, and content organization. In the same context, Sally Sia Yin Yin and M. Mohamad (2023) reported that teachers perceive digital tools as beneficial in developing lesson plans that are more dynamic and engaging, as these tools support the creation of instructional materials that are interactive, resource-rich, and adaptable to learner needs.

In terms of actual implementation in lesson planning workflows, research shows that educators rely on multiple digital platforms to prepare and organize instructional plans. Windi Azahra and M. Muliana (2025) identified commonly used platforms such as Google Classroom, Zoom, and Canva tools that support collaborative lesson design and structured instructional preparation.

Benjamin Luke Moorhouse and Lu Yan (2023) further found that teachers use different categories of digital tools core instructional tools, supplementary applications, and remote learning platforms depending on how frequently and purposefully they are integrated into their teaching and lesson planning practices.

Similarly, Božidar Ilić (2025) observed that teachers actively use tools such as Microsoft Teams, PowerPoint, YouTube, and messaging applications like Messenger or email in preparing and delivering lesson-related content. These findings indicate that while digital tools are widely used in lesson planning, they remain fragmented across multiple platforms, creating inefficiencies in organizing instructional materials.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Despite these benefits, studies also highlight several challenges in the effective implementation of online lesson planning systems. Sally Sia Yin Yin and M. Mohamad (2023) identified barriers such as lack of teacher training, technical difficulties, and challenges in maintaining student engagement when using digital tools for instructional planning.

Božidar Ilić (2025) also reported issues such as increased workload, technical limitations, and unequal access to technology among students, which can affect the consistency of lesson planning and delivery. Additionally, Windi Azahra and M. Muliana (2025) emphasized that gaps in digital literacy and unequal access to technological resources continue to hinder effective integration of digital lesson planning tools.

Academic Resource Management System

The implementation of web-based, centralized academic resource management systems has shown significant potential in improving institutional efficiency, accessibility, and overall quality of educational administration. Prof. Waqar Khan (2026) found that AI-driven resource management systems integrated with centralized databases enhance accessibility, administrative convenience, response time, and system transparency compared to traditional manual processes.

Supporting this, Dr. Varsha Agarwal (2024) reported that effective digital resource management in academic institutions improves resource utilization and



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

accessibility, leading to better organization of electronic materials, databases, and instructional content that ultimately supports improved academic outcomes.

In terms of system implementation, studies show that academic institutions utilize modern web technologies and integrated frameworks to support resource management workflows. Prof. Waqar Khan (2026) identified technologies such as PHP, MySQL, HTML, CSS, JavaScript, AI-driven query processing, and conversational frameworks like Rasa to enhance system interaction and usability.

Similarly, W. Salama and Moustafa H. Aly (2024) demonstrated the use of full-stack web technologies including Node.js, MongoDB, JWT authentication, and responsive design to build school management systems that support real-time communication and multilingual functionality.

In addition, B. Tjiptady (2024) observed that centralized platforms such as Masterweb integrate multiple digital services, enabling seamless access to academic tools and supporting blended learning environments. .

The effectiveness of centralized academic resource management systems is further supported by quantitative evidence demonstrating improvements in usability and institutional performance. Jian Li (2025) reported high system performance metrics, including 92% precision, 88% recall, and strong user satisfaction ratings, indicating that users find centralized academic resource systems efficient and reliable.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Despite these advantages, several challenges continue to affect the successful implementation of academic resource management systems. Dr. Varsha Agarwal (2024) identified barriers such as limited technological competence, low digital awareness, and insufficient training among users, which can hinder system adoption. Furthermore, Putri Pramestiwi Kusumojati and Elis Mediawati (2024) emphasized that many institutions still lack internal systems for proper asset and resource tracking, resulting in difficulties in managing and retrieving academic data efficiently.

Local Literature

Educational Technology in the Philippines

The implementation of educational technology in Philippine schools has shown significant potential in improving learning outcomes, instructional effectiveness, and educational accessibility across diverse subject areas. Mave Drick G. Lucaberte & Ronhick E. Sanchez (2024) found that educational technology integration in Filipino subject instruction significantly enhanced Grade 9 students' learning performance ($p=.002$), with experimental groups achieving higher post-test scores ($M=9.41$; $SD=.31$) compared to control groups ($M=9.12$; $SD=.38$).

Supporting this, Rhandy DG. Almadrones & Frederick G. Tadifa (2024) reported that Physics Education Technology (PhET) simulations improved students' proficiency levels on least-mastered competencies in General Physics 1,



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

strengthening claims that technology can serve as virtual laboratories for enhanced learning experiences. Additionally, Rolyn Joy Campado (2023) demonstrated that assistive technology integration for learners with special educational needs provides reinforcement to learning, satisfaction in teaching, and motivational impact to learners.

In terms of system implementation, studies show that Philippine educational institutions utilize various technological approaches to support curriculum delivery. Rhandy DG. Almadrones & Frederick G. Tadifa (2024) identified PhET simulations as effective tools for demonstrative applications in physics education, while Julius Ceazar G. Tolentino (2024) documented the use of integrated technologies and online education platforms in physical education curriculum implementation.

Furthermore, M. Cadiz (2024) observed through bibliometric analysis of 61 articles that higher education institutions align educational paradigms with Industrial Revolution 4.0 imperatives, incorporating diverse digital technologies to foster academic excellence.

The effectiveness of educational technology integration is supported by quantitative evidence demonstrating improvements in learning outcomes and institutional performance. Analyn M. Gamit (2023) found that teachers from private schools showed higher ICT implementation assessment scores, while younger teachers demonstrated greater effectiveness in ICT-enhanced student learning compared to older counterparts, indicating positive correlations between technology integration and educational outcomes.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Despite these advantages, several challenges continue to affect successful educational technology implementation in Philippine schools. A. H. Mastul (2023) identified barriers including lack of infrastructure, inadequate teacher training, insufficient funding, digital divide disparities, and resistance to change that hinder effective EduTech utilization. Furthermore, Allan Jay Esteban (2024) emphasized that no research exists on digital technology integration for reading instruction in the Philippine context, despite demonstrated efficiency in other countries, resulting in significant gaps in technology-enabled literacy development.

DepEd Digital Learning Systems

DepEd's digital learning systems and the Most Essential Learning Competencies (MELCs)-based curriculum in Philippine schools have shown strong potential in improving instructional delivery and student learning outcomes across different educational contexts. Judy G. Zalun (2023) found that MELCs utilization significantly enhanced Grade Six students' academic development in cognitive, affective, and psychomotor domains through structured lesson alignment, flexible activities, and appropriate assessment strategies.

In addition, Julius Ceazar G. Tolentino (2024) reported that technology integration in curriculum delivery, particularly in Physical Education, supported continuous learning through the use of online platforms and digital tools, especially during the COVID-19 pandemic. Dewiapriani Fristianingroem (2025) further emphasized that Learning Management System (LMS)-based instruction improves



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

educational effectiveness by promoting flexibility, efficiency, and alignment with workforce-oriented competencies through project-based learning approaches.

In Philippine school settings, educators have been observed using various digital tools to support MELCs-based curriculum delivery and instructional planning. Babylyn Javier (2022) noted that teachers demonstrate moderate use of digital teaching and learning technologies, reflecting both growing adoption and existing limitations in technical readiness and implementation consistency. Tomas Jr Diquito (2024) also highlighted that the MATATAG curriculum integrates technology and engineering literacy, aligning educational competencies with the demands of Industry 4.0 and Industry 5.0. These observations suggest that while digital integration in curriculum delivery is progressing, its application remains uneven and heavily dependent on teacher capability and institutional support, indicating the need for a more structured and centralized system to assist in instructional planning.

Findings on curriculum effectiveness show that MELCs-based implementation is generally positively received by educators but still requires further strengthening. D. Montebon (2024) reported moderate levels of agreement among teachers regarding MELCs implementation in terms of instructional impression, morale, and workload management, suggesting that while the framework is functional, its full optimization in classroom practice is still developing. This implies that improvements in instructional support systems are still necessary to maximize curriculum effectiveness in daily teaching practice.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Despite these developments, several challenges continue to hinder effective implementation of MELCs and digital learning systems in Philippine schools. Reyner Villaseñor (2025) identified barriers such as poor internet connectivity, lack of devices, limited school resources, and socio-economic disparities that contribute to the digital divide. Similarly, Judy G. Zalun (2023) pointed out additional challenges including limited parental and student cooperation, insufficient financial resources, and lack of instructional materials that affect curriculum delivery. These issues highlight that while DepEd's digital learning initiatives and MELCs framework provide a strong foundation for modern education, their effectiveness is constrained by infrastructural and implementation challenges.

Lesson Planning Challenges of Filipino Teachers

Lesson planning difficulties and administrative workload intensification among Filipino teachers have been identified as significant factors affecting instructional quality and teacher well-being in Philippine educational contexts. Paolo Niño Valdez et al. (2024) found that English language teachers exhibit differing perceptions of lesson planning, with some viewing it as a creative and reflective process while others consider it routine and repetitive, creating inconsistencies that complicate effective professional development design.

Similarly, Michael T. Sebullen (2023) reported that pre-service teachers experience challenges in formulating clear instructional objectives, aligning learning outcomes with activities and assessments, and integrating ICT due to limited



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

equipment and insufficient technical skills among both learners and cooperating teachers.

In addition to planning challenges, administrative workload has been consistently identified as a major factor contributing to teacher stress and reduced instructional effectiveness. Zusette Orena and Lorelie Gamutan (2025) found significant relationships between workload demands and teacher performance across ancillary services, classroom tasks, administrative duties, and research responsibilities, indicating that multiple tasks demand negatively affect teaching efficiency.

Supporting this, Elizabeth L. Lagawid (2024) emphasized that administrative responsibilities combined with lesson planning contribute to “time poverty,” limiting teachers’ capacity to focus on effective pedagogy and professional growth.

Despite these challenges, teachers demonstrate adaptability through various coping strategies to manage their responsibilities. Emely E. Lacorte (2025) found that teachers apply structured scheduling, task prioritization, and time blocking techniques to handle their workloads effectively. In addition, Marie Feel Cornista (2025) reported that educators use time management, collaboration, and resourcefulness as coping mechanisms in response to administrative task demands, although these strategies remain insufficient due to limited institutional support and inconsistent policy implementation. These findings indicate that while teachers actively develop personal strategies to manage workload pressures, systemic support is still lacking, highlighting the need for institutionalized solutions.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

These issues collectively demonstrate that lesson planning challenges and administrative workload concerns are interconnected systemic problems within the Philippine education system. Ederliza D. Frianeza et al. (2024) emphasized that broader structural issues such as limited government funding, curriculum implementation constraints, and insufficient institutional support significantly affect instructional quality. While teachers show resilience in managing their responsibilities, the persistence of these challenges suggests that sustainable improvement requires structural interventions rather than individual coping alone.

Philippines Academic Resource / School Systems

Web-based academic resource management systems and school management systems in the Philippines have demonstrated strong potential in improving educational administration efficiency, although their implementation is still affected by technological, infrastructural, and capacity-related constraints. Studies show that web-based school management information systems significantly enhance administrative processes by improving data handling across student records, teacher information, financial transactions, and reporting systems. These systems also contribute to greater transparency, faster information access, and improved accuracy in school operations, highlighting their value in modernizing educational administration Eka Indah Wahyuni et al. (2025).

In a practical Philippine context, digital document management systems have also been successfully implemented, allowing school administrators to digitize



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

traditional paper-based processes and enabling online posting, access, downloading, and submission of official documents, thereby improving the efficiency of dissemination and collection of institutional requirements, Niño Nagrama. (2024).

Despite these benefits, several studies highlight persistent challenges in the implementation of web-based management systems in Philippine educational institutions. Findings from higher education resource management indicate that while budgeting and allocation processes are consistently implemented, planning and control mechanisms are only moderately practiced, reflecting uneven adoption of systematic management approaches Josephine V. Acido & Dennis Anthony A. Kilongkilong, (2022).

In addition, evidence shows that when properly implemented, web-based management systems contribute positively to educational outcomes in Philippine schools. A large-scale study involving public schools revealed a significant positive correlation between school-based management practices and academic achievement, with improvements observed in student performance and attendance in institutions that adopt structured management systems Jason Ang et al. (2025). These findings indicate that systematic digital management approaches can enhance both administrative efficiency and educational performance, provided that implementation challenges are addressed effectively.

Foreign Studies

AI in Education System Studies



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

AI implementation in educational systems has demonstrated significant potential for enhancing teaching effectiveness and student learning outcomes, though implementation faces substantial challenges requiring systematic support. Muqorobin (2025) conducted a mixed-methods study across four countries (Indonesia, India, Finland, and the United States) and found that AI technologies—including intelligent tutoring systems, predictive analytics platforms, natural language processing, and automated assessment systems—contribute positively to personalized learning experiences, enhance assessment and feedback efficiency, and aid in streamlining school administration.

Similarly, Sanket Napte & D. Singh (2025) reported through experimental design that AI implementation resulted in 20% improvement in retention and test scores compared to control groups, with increased student engagement through customized content delivery and reduced administrative workload for teachers.

The effectiveness of AI-based learning systems is particularly evident in personalized learning capabilities. Ettyani (2025) found through systematic literature review that AI enables personalized learning by adapting materials, methods, and pace according to each student's needs, with technologies such as intelligent tutoring systems, learning analytics, and educational chatbots improving learning experiences and increasing student engagement and independence through adaptive feedback.

Cindy Fitri Laksono et al., (2024) confirmed through qualitative research that AI facilitates personalized learning experiences by analyzing individual student data



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

to tailor educational content according to their unique strengths and weaknesses, enhancing teaching efficiency by automating administrative tasks like grading and attendance tracking.

Teacher perspectives and adaptation strategies show promise for successful implementation. Norbert Annuš (2024) conducted a qualitative survey involving teachers and found that educators are open to AI technologies regardless of generation and discipline, emphasizing that effective education requires appropriate use of these technologies while maintaining teacher attention to students and active communication.

Cindy Fitri Laksono et al., (2024) found that AI-based training programs, such as AloT-PD (AI of Things-Professional Development), successfully improve teachers' competency in AI, demonstrating that technology can be an effective tool to support lifelong learning and help teachers develop new skills throughout their careers.

AI-Assisted Lesson Planning Studies

AI-assisted lesson planning systems have demonstrated strong potential in reducing teacher workload and improving instructional quality, making them increasingly relevant in modern educational environments. Julie Carson et al. (2026) found that the use of AI tools such as ChatGPT led to a significant 52.5%



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

reduction in weekly lesson planning time, while also increasing teacher confidence in planning tasks.

Similarly, Deepak Varuvel Dennison et al. (2025) reported that AI-assisted planning helps reduce bureaucratic workload, minimize planning time, and lower teaching-related stress, while encouraging more activity-based and student-centered instructional approaches.

The effectiveness of AI in lesson planning is further evident in its ability to support differentiated instruction and improve administrative efficiency. K. Karpouzis et al. (2024) found that generative AI tools enhance learning experiences by accommodating diverse student needs while simultaneously reducing the time required for lesson preparation.

In addition, Sabrina Belloula (2025) reported that AI tools enable teachers to focus more on creative instructional activities by reducing time spent on routine administrative tasks, resulting in improved lesson quality and increased student engagement.

Teacher perspectives on AI integration highlight both readiness and limitations in practical application. Dini Dwi Rahayu et al. (2025) found that teachers generally exhibit positive attitudes toward AI tools, perceiving them as useful, easy to use, and beneficial for instructional tasks.

However, Endang Setyaningsih et al. (2024) emphasized that while AI accelerates lesson planning and supports differentiated instruction, its outputs often



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

require refinement to ensure pedagogical appropriateness and classroom readiness.

Learning Management System (LMS) Studies

Learning Management Systems (LMS) have demonstrated strong potential in improving educational delivery, student engagement, and institutional efficiency, making them a core component of modern higher education environments. E. D. Madyatmadja et al. (2023) found that web-based LMS platforms significantly enhance linkage between instructional activities, improve system effectiveness, and increase student satisfaction in higher education settings while also streamlining administrative processes and improving overall learning experiences.

Similarly, Ani Herniawati et al. (2025) reported through systematic review that LMS platforms play a key role in increasing student engagement, enhancing learning motivation, and improving overall educational quality, particularly in online and blended learning environments.

The effectiveness of LMS platforms is further demonstrated through their ability to support personalized learning and improve institutional efficiency. Ani Herniawati et al. (2025) emphasized that LMS platforms enable flexible learning pathways, personalized content delivery, and performance tracking through learning analytics, allowing educators to better monitor and support student progress. In addition, L. Sánchez et al. (2024) identified high-performing LMS



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

platforms such as Paradiso and Moodle, which achieved strong evaluation scores and demonstrated widespread global adoption, with Moodle alone serving over 250 million users.

User perspectives on LMS adoption reveal generally positive acceptance, alongside implementation challenges that affect full utilization. Ahlam I. Almusharraf (2024) found that ease of use is a major factor influencing student acceptance of LMS platforms, along with interactivity and relevance of learning content. However, Ani Herniawati et al. (2025) also highlighted persistent challenges including unequal access to technology, resistance to pedagogical change, and infrastructure limitations, particularly in underdeveloped regions.

Web-Based Education System Case Studies

Web-based education systems have demonstrated strong potential in improving educational delivery, student engagement, and institutional efficiency, making them a core component of modern school environments. M. Saracostti et al. (2022) found that web-based systems significantly enhance decision-making processes between teachers, students, and parents while streamlining school management and supporting day-to-day educational practices through collaborative development between universities, IT teams, and school communities.

Similarly, Listia Utami et al. (2024) reported exceptional user satisfaction with an average score of 99% and overwhelmingly “strongly agreed” responses across



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

all evaluation metrics, demonstrating the system's effectiveness in enhancing learning efficiency and streamlining information management within school environments.

The effectiveness of web-based education systems is further demonstrated through their ability to support comprehensive educational management and improve institutional processes. Dewi Yulianti et al. (2025) emphasized that digital technology in learning increases student interest, motivation, and understanding while making the learning process more interactive and aligned with technological developments in the digital era.

In addition, Igor Balaban & Aleksandra Sobodić (2022) demonstrated large-scale viability through successful implementation across six European countries involving more than 800 teachers and 6,000 students, confirming that effective integration and evaluation within formal curricula in primary and secondary schools is achievable.

User perspectives on web-based education system adoption reveal generally positive acceptance, alongside implementation challenges that affect full utilization. P. Nugroho & Safuan Safuan (2025) found that while web-based platforms are widely adopted, their effectiveness depends on infrastructure readiness and digital literacy of educators and students.

However, Win Min Zaw & S. S. Hlaing (2024) also highlighted persistent challenges including teacher shortages, limited resources, and geographic



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

isolation, particularly in developing countries, though proper infrastructure and training support can maximize educational equity potential.

Educational Technology Impact Studies

Educational technology has consistently demonstrated strong potential in enhancing student learning outcomes across diverse educational contexts, establishing itself as a key driver of transformation in modern education systems. Xiu-Yi Wu (2023), through a comprehensive meta-analysis of 60 peer-reviewed studies, found that digital technology positively influences deep learning outcomes, with particularly strong effects in humanities and social science disciplines.

Similarly, Olateju Temitope Akintayo et al. (2024) synthesized findings from 47 empirical studies and reported that educational technology improves student engagement, enhances knowledge retention, and supports the development of higher-order thinking skills when properly integrated into teaching and learning processes.

The effectiveness of educational technology is further supported by large-scale evidence showing measurable improvements in academic performance and instructional outcomes. Jesús Valverde-Berrocoso et al. (2022) analyzed 100 research articles and found consistent evidence that ICT-enhanced educational



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

practices improve student performance, particularly in mathematics and science education where technology integration is most extensively studied.

In addition, S. Aljehani (2024) confirmed through a quantitative study involving 394 participants that technology-enhanced instruction significantly improves both academic performance and student engagement while accommodating diverse learning styles.

From an implementation perspective, research emphasizes that the effectiveness of educational technology depends heavily on pedagogical alignment and appropriate integration strategies. Xiu-Yi Wu (2023) found that blended approaches combining online and offline learning environments produce more significant improvements in deep learning outcomes compared to purely online implementations, highlighting the importance of structured and balanced integration.

Similarly, Xianhua Zhao Ma et al. (2024) observed that technology integration enhances student engagement and skill development, particularly in critical thinking, problem-solving, and collaboration, when supported by appropriate instructional design.

Despite these benefits, studies consistently highlight challenges that limit the full effectiveness of educational technology in practice. Olateju Temitope Akintayo et al. (2024) emphasized that the impact of technology varies depending on the type of tool used, pedagogical approach, and implementation context, underscoring the importance of teacher training and institutional support.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

In addition, Rica Arisanti (2025) noted that although digital technologies such as multimedia tools, e-learning platforms, and augmented reality offer significant educational benefits, their effectiveness depends on teacher competency, infrastructure quality, and instructional design.

Local Studies

Educational Technology Philippines Study

Educational technology research in the Philippines highlights its strong potential to improve learning outcomes, although widespread effectiveness remains limited due to systemic implementation challenges. Czarina Enorme et al. (2024), through a systematic literature review of selected studies from a large pool of scholarly articles, found that ICT integration in mathematics education enhances student engagement, supports innovative pedagogical strategies, improves classroom management, and strengthens assessment practices. The study further emphasized that ICT enables computer-supported collaborative learning through the use of virtual manipulatives and simulations that help develop higher-order thinking skills.

In addition, Monica Sophia Reyes and Joy San Buenaventura-Gaza (2025) reported that elementary teachers in the Philippines are able to implement technology-based instructional practices aligned with global standards, reflecting



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

the country's capacity to adopt effective educational technologies when proper conditions are present.

From an implementation perspective, Philippine studies emphasize the importance of institutional support and structured frameworks in achieving successful technology integration. Amanda Jane Menta (2022) demonstrated that applying the Technology–Environment–Organization (TEO) framework in higher education institutions can streamline operations, reduce IT-related costs, improve system scalability, and enhance collaboration efficiency. The study also provided strategic recommendations for national-level implementation, particularly for the Commission on Higher Education, highlighting the need for systematic and policy-driven digital transformation across institutions.

Despite these advantages, Philippine-based research consistently identifies critical barriers that hinder effective educational technology integration. Czarina Enorme et al. (2024) pointed out that many schools lack adequate ICT infrastructure, including reliable internet access and sufficient devices, while financial limitations restrict the acquisition of necessary technologies.

Additionally, gaps in teacher training, limited technical support, and low confidence in using digital tools contribute to inconsistent implementation across classrooms. Allan Jay Esteban et al. (2024) further identified a significant research gap, noting the absence of studies on digital technology integration for reading instruction in the Philippine context despite strong international evidence of its effectiveness. Moreover, Sali S. Mannan et al. (2025) highlighted socio-cultural and



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

economic barriers, including security concerns and limited access to ICT resources, which particularly affect participation in technology programs in certain regions.

DepEd Online Learning Effectiveness Study Philippines

DepEd's distance and online learning implementation in the Philippines has shown measurable improvements in student academic performance, although its effectiveness varies depending on the learning modality and the level of institutional support provided. Jonathan Roque (2022), through a systematic review of educational experiences during the pandemic, found that the Department of Education's flexible blended learning model resulted in significant improvement in students' knowledge of the specific content covered, despite the presence of technical and non-technical challenges during implementation.

Supporting this, M. Gracia et al. (2022) provided empirical evidence of successful online learning implementation in a private school setting, involving 1,808 participants composed of students, faculty, and administrators. The study reported that learners achieved above-passing performance levels with highly satisfactory final grades during full online learning implementation, demonstrating that positive academic outcomes are achievable when adequate systems and conditions are in place.

From a systemic perspective, Philippine research highlights that the success of distance learning depends heavily on the presence of comprehensive support



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

structures. J. C. Tanucan et al. (2023), through meta-synthesis analysis, identified four essential components of effective modular distance learning: structural presence, community presence, intervention presence, and quality assurance presence. These components emphasize that effective learning delivery requires coordinated efforts among instructional design, stakeholder collaboration, learner support systems, and continuous evaluation processes, indicating the need for well-organized and structured educational systems.

Despite these positive outcomes, significant implementation challenges continue to affect the effectiveness of distance learning in the Philippines. Dave E. Marcial et al. (2024) found that although educators demonstrate high motivation toward improving online teaching practices, they exhibit deficiencies in actual online learning competencies, revealing a critical gap between willingness and practical skill application.

Technology Integration Philippine Schools Research

ICT integration in Philippine schools demonstrates strong potential for improving educational outcomes, with teachers generally exhibiting high technological competence; however, its effectiveness remains constrained by infrastructure limitations and uneven institutional support. Thea Marcel Portugues (2026), through a study involving 344 teachers in the Schools Division of Antique, found that educators demonstrated high levels of ICT proficiency, particularly in software use and digital literacy, with strong integration practices observed in



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

instructional planning, lesson delivery, and assessment. The study further revealed a significant positive relationship between teachers' ICT competence and learners' academic performance, indicating that teacher technological proficiency plays a critical role in improving educational outcomes.

Supporting this, Norlie L. Cañete (2026) found that while junior high school teachers in Northern Negros generally experienced low difficulty in technical ICT skills, they still actively employed mitigation strategies to address implementation challenges. These strategies included self-directed learning, peer collaboration, and adaptive instructional practices, reflecting teachers' efforts to continuously adjust and improve technology integration despite contextual limitations. This suggests that Filipino educators are not passive adopters of technology but active problem-solvers in addressing implementation gaps.

From a disciplinary perspective, ICT integration shows varying levels of effectiveness across subject areas in the Philippine context. Czarina Enorme et al. (2024), through a systematic review of mathematics education studies, found that ICT use significantly enhances student engagement, improves classroom management, strengthens assessment practices, and supports computer-assisted collaborative learning through tools such as virtual manipulatives and simulations. These tools also contribute to the development of higher-order thinking skills, demonstrating that ICT is particularly effective when aligned with subject-specific pedagogical strategies.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Despite these positive developments, significant implementation gaps continue to limit the full effectiveness of ICT integration in Philippine education. Allan Jay Esteban et al. (2024) identified a critical research gap, noting the absence of studies on digital technology integration for reading instruction in the Philippine context, despite strong international evidence supporting its effectiveness in literacy development.

In addition, Czarina Enorme et al. (2024) highlighted persistent systemic barriers including inadequate ICT infrastructure, unreliable internet connectivity, insufficient digital devices, budget constraints, and limited technical support, all of which contribute to inconsistent ICT utilization across schools. These challenges indicate that while ICT competence among teachers is strong, the lack of supporting systems and infrastructure prevents full optimization of technology in education.

Learning Management System Philippines Study

Learning Management System (LMS) implementation in educational institutions demonstrates strong potential for improving learning effectiveness and student engagement; however, its success is often constrained by challenges related to digital literacy, infrastructure limitations, and institutional readiness. M. Furqon et al. (2023) found that LMS utilization has a positive effect on students' academic performance and contributes to favorable perceptions of LMS use in academic settings, indicating that LMS platforms can significantly enhance educational outcomes when properly implemented.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Supporting this, Novira Aulianda et al. (2023) identified four key LMS implementation stages introduction, registration, learning materials, and evaluation and found that core LMS features such as communication tools, course content delivery, and assignment management significantly influence student satisfaction, engagement, motivation, learning effectiveness, and overall academic performance.

Similarly, Rekha A P (2024), through a systematic review, confirmed that LMS platforms improve student learning outcomes and enhance engagement and satisfaction, although their effectiveness depends on instructional design, LMS type, and the level of student participation.

Despite these benefits, substantial barriers continue to limit the full effectiveness of LMS implementation across educational contexts. Silvia P Ningsih et al. (2025) identified major challenges such as low digital literacy among educators, inadequate technological infrastructure, and limited user readiness in higher education institutions.

In addition, A. Aziz et al. (2024) highlighted that although LMS adoption is generally accepted by students, concerns regarding privacy and data security affect user trust, emphasizing the importance of secure and reliable system design.

These challenges are more pronounced in developing countries, where LMS adoption requires stronger institutional support and policy direction. Dewiapriani Fristianingroem (2025) emphasized that successful LMS implementation depends on improving technological infrastructure, providing continuous training for



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

educators, ensuring technical support, developing interactive learning content, and strengthening funding and policy frameworks.

Student Performance in Digital Learning in Philippines

Digital learning technologies demonstrate strong potential for improving student performance and engagement across educational contexts; however, their effectiveness is highly dependent on implementation quality, institutional support, and contextual conditions. Jesús Valverde-Berrocso et al. (2022) found consistent evidence that ICT-enhanced educational practices improve academic performance, particularly in mathematics and science education, while Ashraf Bany Mohammed et al. (2024) demonstrated that task-technology fit and student satisfaction significantly contribute to learning performance and overall academic success in online learning environments. These findings indicate that digital learning tools can enhance educational outcomes when properly aligned with instructional design and learner needs.

Supporting this, Muhammad Rafiq-uz-Zaman (2023) confirmed that digital literacy significantly improves learning outcomes when combined with effective instructional design, institutional support, and metacognitive learning strategies, leading to improvements in academic performance, engagement, and higher-order thinking skills.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Similarly, Zhonggen Yu (2022) found that the integration of digital technologies with appropriate teaching strategies, teacher-student collaboration, social regulation, and gamification can significantly enhance online learning outcomes, particularly during emergency remote learning situations such as the COVID-19 pandemic. These studies suggest that the effectiveness of digital learning systems depends not only on technology availability but also on pedagogical alignment and user competence.

Despite these advantages, research consistently identifies significant challenges that limit the effectiveness of digital learning technologies. C. Akpen et al. (2024) reported mixed outcomes of online learning, including reduced engagement, learner isolation, and limited interaction between instructors and peers, with effectiveness further constrained by poor internet connectivity, low-quality digital tools, and reduced student motivation.

In addition, Stella Timotheou et al. (2022) highlighted that many educational institutions experienced low digital capacity and limited experience during the COVID-19 transition, resulting in widened educational inequalities and learning losses.

Collectively, these findings emphasize that digital learning transformation is a complex process influenced by multiple interrelated factors. Muhammad Rafiq-uz-Zaman (2023) further noted that the impact of digital learning technologies is inconsistent and strongly mediated by access, pedagogical design, and educator



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

readiness, highlighting the need for more holistic and context-sensitive implementation approaches.

Synthesis of the Reviewed Literature and Studies

Across the reviewed foreign and local literature, educational technology consistently demonstrates strong potential to improve teaching efficiency, learning outcomes, and institutional management through tools such as artificial intelligence, machine learning, learning management systems, ICT integration, and web-based platforms. Studies show that these technologies enhance instructional delivery by reducing teachers' administrative workload, supporting lesson planning, and centralizing academic resources, while also improving student engagement, motivation, and academic performance when properly implemented.

However, the literature also emphasizes that technology alone is not sufficient to guarantee success, as teachers remain central in ensuring meaningful and context-appropriate use of digital tools. Across both global and Philippine contexts, persistent challenges such as inadequate infrastructure, limited digital literacy, insufficient training, unstable internet connectivity, and resource constraints continue to hinder full implementation and effectiveness.

Overall, the findings suggest that educational technology can significantly transform education, but its success depends on a balanced integration of technology, teacher capability, and strong institutional support systems.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

CHAPTER 3 METHODOLOGY

Research Design

This study will utilize the developmental-descriptive research design in the design, development, implementation, and evaluation of the EduScribe system. Developmental research focuses on the creation, improvement, and validation of systems or solutions intended to address existing problems within a specific environment. In this study, the approach will be applied in developing an AI-assisted web-based lesson planning and academic resource management system for Dela Paz National High School.

The descriptive method will be used to evaluate the effectiveness, usability, and acceptability of the developed system among its intended users. It will involve gathering feedback from teachers and students regarding the system's functionality, efficiency, reliability, and overall performance. Through this approach, the researchers will determine whether the EduScribe system effectively addresses the challenges encountered in lesson planning, academic resource management, assessments, and communication processes.

Furthermore, the combination of developmental and descriptive research will enable the researchers to systematically develop the system while simultaneously evaluating its effectiveness and responsiveness to user needs. This research design



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

is considered appropriate because it supports both the development of the EduScribe system and the assessment of its performance and acceptability among intended users.

Software Development Methodology

The study will utilize the Modified Waterfall Model as the software development methodology in designing and developing the EduScribe system. The Modified Waterfall Model is a structured software development approach that follows sequential phases while allowing limited revisions and feedback between stages whenever necessary. This methodology is considered appropriate for the study because it provides a systematic and organized process in developing academic systems while ensuring proper planning, documentation, testing, and evaluation.

Unlike the traditional Waterfall Model, the Modified Waterfall Model allows feedback loops between phases to improve system quality and address issues identified during development. The methodology will ensure that each phase of the project is completed and evaluated before proceeding to the next stage, minimizing errors and maintaining consistency throughout the development process.

The Modified Waterfall Model to be used in the development of the EduScribe system consists of the following phases: Requirement Gathering, System Design, Implementation, Testing, and Deployment and Maintenance. This methodology will

enable the researchers to develop the system in a structured and organized manner while still allowing improvements throughout the development process.

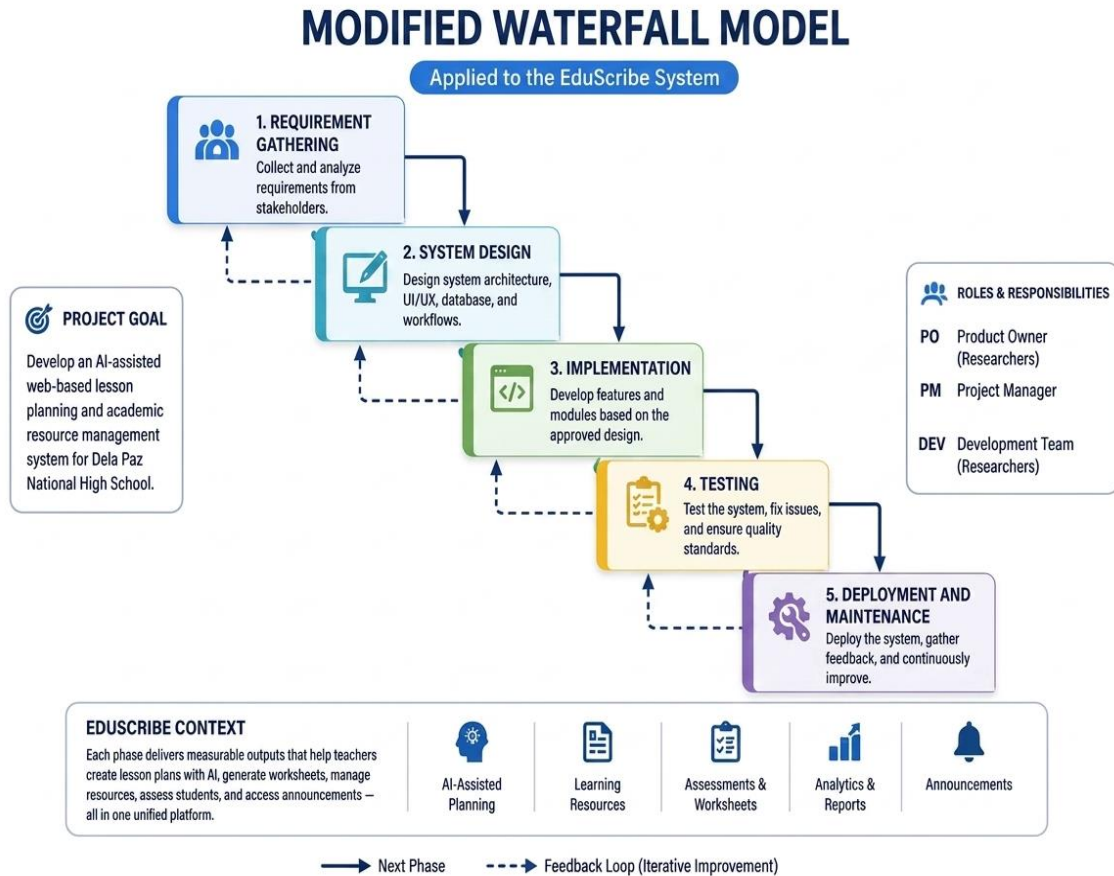


Figure 3.1 Modified Waterfall Model Applied to the EduScribe System

Figure 3.1 presents the Modified Waterfall Model to be applied in the development of the EduScribe system. The process begins with the project goal of developing an AI-assisted web-based lesson planning and academic resource management system for Dela Paz National High School.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Requirement Gathering. The first phase will involve collecting and analyzing the requirements of the system from the intended users and stakeholders. The researchers will identify the problems encountered by teachers and students in lesson planning, academic resource management, assessments, and communication processes. Relevant information will be gathered through surveys, interviews, observations, and related literature to determine the features and functionalities needed in the system.

System Design. After gathering the requirements, the researchers will proceed to the system design phase. This phase will focus on designing the overall architecture of the EduScribe system, including the database structure, user interface design, system workflows, and navigation processes. The researchers will also plan the integration of artificial intelligence features and security measures such as user authentication and access control.

Implementation. In this phase, the researchers will develop and code the system based on the approved system design. The major features to be implemented in the EduScribe system include AI-assisted lesson planning, learning resource management, assessments and worksheets, analytics and reports, and announcement management. The researchers will utilize appropriate programming languages, frameworks, and development tools to ensure proper system functionality and performance.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Testing. The testing phase will involve evaluating the developed system to identify errors, bugs, and usability issues. The researchers will conduct functional testing, usability testing, and performance evaluation to ensure that the EduScribe system operates accurately, efficiently, and reliably. Necessary revisions and improvements will be implemented whenever issues are identified during testing.

Deployment and Maintenance. After successful testing, the EduScribe system will be deployed for actual usage and evaluation by the intended users. During this phase, the researchers will monitor the system's performance and gather feedback from teachers and students regarding the functionality and effectiveness of the platform. Maintenance activities such as system updates, issue resolution, and continuous improvements will also be performed to ensure long-term usability and reliability of the system.

Feedback Loop (Iterative Improvement). Although the methodology follows a sequential process, the Modified Waterfall Model allows feedback loops between phases whenever revisions or improvements are necessary. This iterative improvement process will enable the researchers to revisit previous phases, particularly during testing and evaluation, to refine system functionalities and improve overall system quality.

System Development Tools and Technologies. The researchers will utilize several technologies and development tools in building the EduScribe system. These



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

technologies will support frontend and backend development, database management, artificial intelligence integration, authentication, and system deployment.

Component	Technology Used	Purpose
Frontend Development	React.js	Used for developing the user interface
Backend Development	Node.js	Used for server-side processing and API handling
Database Management	Supabase PostgreSQL	Used for storing and managing system data
Authentication	Supabase Authentication	Used for secure login and user access management
Artificial Intelligence	Google Gemini API	Will be used for AI-assisted lesson planning and content generation
Styling Framework	Tailwind CSS	Used for responsive and modern interface design
Version Control	GitHub	Used for project collaboration and source code management
Deployment	Vercel / Render	Used for hosting and deploying the system

Source of Data



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

The data to be used in this study will be obtained from both primary and secondary sources.

Primary Source. Primary data will be collected from selected teachers and students of Dela Paz National High School, who will serve as the main respondents of the study. These individuals are the primary users of the EduScribe system, as they are directly involved in lesson planning and academic resource management. Data will be gathered through surveys and questionnaires to identify their experiences, challenges, and feedback regarding the current process and the proposed system.

Secondary Source. Secondary data will be obtained from related literature, academic studies, online resources, journals, books, and official documents such as the Most Essential Learning Competencies (MELC). These sources will provide supporting information for the study and help ensure that the system aligns with educational standards and existing research related to academic planning and resource management.

Research Instrument

The researchers will utilize various instruments to gather appropriate and relevant data needed for the design, development, implementation, and evaluation of the EduScribe system. These instruments include online research, surveys, and questionnaires, which will be used to collect information essential for system development and assessment.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Online Research. Online research will involve gathering information from academic articles, journals, websites, and other credible online materials. This method will allow the researchers to collect related literature, references, and existing studies relevant to artificial intelligence integration, lesson planning systems, and academic resource management.

Survey and Questionnaires. Survey questionnaires will be utilized to systematically collect feedback from respondents who will interact with the EduScribe system. The questionnaire is designed to evaluate the overall performance and acceptability of the EduScribe system in terms of usability, functionality, efficiency, and reliability. These criteria will be used to determine whether the system effectively meets the needs of its intended users and provides satisfactory user experience and system performance.

A five-point Likert Scale will be used to measure the respondents' level of agreement regarding the different evaluation criteria of the system.

Data Gathering Procedure

The researchers will first secure permission and approval from the administration of Dela Paz National High School before conducting the study and distributing the survey questionnaires.

After approval is granted, the researchers will coordinate with selected teachers and students who will serve as respondents of the study. The researchers will prepare and validate the survey questionnaires based on the objectives of the



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

research. The questionnaires will be converted into Google Forms for easier distribution and data collection.

The survey link will be distributed through online communication platforms. Respondents will be informed about the purpose of the study and will be requested to answer the questionnaires honestly and voluntarily.

After collecting the responses, the researchers will organize, tabulate, analyze, and interpret the gathered data using appropriate statistical tools such as percentage distribution and weighted mean. The results will be used to evaluate the acceptability, effectiveness, and overall performance of the EduScribe system. In addition, online research and review of related literature will be conducted to support the development and improvement of the system.

Statistical Treatment of Data

Statistical treatment of data refers to the process of organizing, analyzing, and interpreting collected data in order to convert raw responses into meaningful information. In this study, the data gathered from the survey questionnaires will be analyzed using appropriate statistical tools to evaluate the acceptability and performance of the EduScribe system.

Likert Scale. A Likert Scale is a five-point scale used to measure the attitudes, opinions, and perceptions of respondents ranging from Strongly Agree to Strongly Disagree. In this study, it will be used to gather the responses of the participants regarding the usability, functionality, efficiency, and reliability of the system.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

The five-point Likert Scale will specifically be used to measure the respondents' level of agreement regarding the system's usability, functionality, efficiency, and reliability. The scale will help the researchers collect structured responses that will later be used for quantitative analysis of the system evaluation.

Scale	Description
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

Percentage Distribution

Percentage distribution will be used to determine the proportion of responses gathered from the respondents in relation to the total number of participants.

Formula:

$$\% = \frac{f}{N} \times 100$$

Where:

- f = frequency



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- N = total number of respondents

Weighted Mean. Weighted mean will be used to determine the average evaluation and overall acceptability level of the EduScribe system based on respondents' ratings.

Formula:

$$\bar{x} = \frac{\sum X}{N}$$

Where:

- $\bar{x}$ – weighted mean or average score of the responses
- $\sum X$ – sum of all responses or values obtained from the survey
- N – total number of respondents

This statistical tool will be used by the researchers to compute the average evaluation of each criterion in the study. The results will help interpret the performance and acceptability of the EduScribe system in terms of usability, functionality, efficiency, and reliability.

Rubric Scale Rating

Rubric Scale Rating will be used to interpret the numerical responses gathered from the respondents based on the Likert Scale. This will allow the



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

researchers to classify the results according to their corresponding interpretation levels such as Very High, High, Moderate, Low, and Very Low.

Scale	Description	Interpretation Level
5	Strongly Agree	Very High / Highly Evident
4	Agree	High / Evident
3	Neutral	Moderate / Fair
2	Disagree	Low / Slightly Evident
1	Strongly Disagree	Very Low / Not Evident

Table 3.1

Rubric Scale Rating

These statistical tools will be used to interpret the collected data and evaluate the effectiveness and acceptability of the EduScribe system.

Ethical Considerations



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

The researchers will ensure that ethical standards are strictly observed throughout the conduct of the study. Participation of respondents will be voluntary, and respondents will be informed about the objectives and purpose of the research before data gathering.

Confidentiality and privacy will be maintained by ensuring that personal information will not be disclosed and will be used solely for academic purposes. All gathered data will be securely stored and protected from unauthorized access.

The researchers will comply with the provisions of Republic Act No. 10173, also known as the Data Privacy Act of 2012, by ensuring that all personal information gathered from respondents is treated with strict confidentiality.

Informed consent will be obtained from all respondents, ensuring their willingness to participate in the study. Participants will also be given the right to withdraw from the study at any time without penalty.

Furthermore, the EduScribe system will be designed with security measures such as user authentication and access control to protect sensitive academic information. AI-generated outputs will be reviewed and validated by teachers to ensure accuracy and appropriateness.

The study will also maintain academic integrity by ensuring originality, proper citation of references, and avoidance of plagiarism. Ethical principles such as fairness, inclusivity, and responsible use of technology will be observed throughout the design, development, implementation, and evaluation of the EduScribe system.



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

REFERENCES

- Acido, J. V., & Kilongkilong, D. a. A. (2022). Resource management practices towards sustainable support system during pandemic. *International Journal of Educational Management and Development Studies*, 3(4), 19–42. <https://doi.org/10.53378/352930>
- Agarwal, V., Kaur, S., N, R., Gupta, S., Zaveri, B., Sohal, J., & Shrivastav, A. K. (2024). Adoption of advanced systems for digital resource management in academic institutions. *Evolutionary Studies in Imaginative Culture*, 913–923. <https://doi.org/10.70082/esiculture.vi.1089>
- Akintayo, O. T., Eden, C. A., Ayeni, O. O., & Onyebuchi, N. C. (2024). Evaluating the impact of educational technology on learning outcomes in the higher education sector: A systematic review. *Open Access Research Journal of Multidisciplinary Studies*, 7(2), 052–072. <https://doi.org/10.53022/oarjms.2024.7.2.0026>
- Akpen, C. N., Asaolu, S., Atobatele, S., Okagbue, H., & Sampson, S. (2024). Impact of online learning on student's performance and engagement: a systematic review. *Discover Education*, 3(1). <https://doi.org/10.1007/s44217-024-00253-0>
- AlAli, R., Wardat, Y., Al-Saud, K., & Alhayek, K. A. (2024). Generative AI in education: Best practices for successful implementation. *International Journal of Religion*, 5(9), 1016–1025. <https://doi.org/10.61707/pkwb8402>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Almadrones, R. D., Tadifa, F. G., & La Union Philippines Ftadifa@dmmsuEduPh, D. S. A. (2024). Physics Educational Technology (PHET) Simulations in Teaching General Physics 1. *International Journal of Instruction*, 17(3), 635–650. <https://doi.org/10.29333/iji.2024.17335a>
- Almusharraf, A. I. (2024). An Investigation of University Students' Perceptions of Learning Management Systems: Insights for Enhancing Usability and Engagement. *Sustainability*, 16(22), 10037. <https://doi.org/10.3390/su162210037>
- Ang, J. V., Adlit, M. F., Docdoc, V. A., Comeros, J. S., & Lagnason-Pacheco, L. (2025). School-based management implementation and educational outcomes in Philippine public schools: An explanatory sequential inquiry. *Edelweiss Applied Science and Technology*, 9(4), 2625–2644. <https://doi.org/10.55214/25768484.v9i4.6611>
- Annuš, N. (2024). Education in the age of artificial intelligence. *TEM Journal*, 13(1), 404–413. <https://doi.org/10.18421/TEM131-42>
- Arisanti, R. (2025). The Role of Digital Technology in Improving Learning Outcomes: A Literature-Based Analysis. *PPSDP International Journal of Education*, 4(2), 574–581. <https://doi.org/10.59175/pijed.v4i2.541>
- Aulianda, N., Wijayati, P. H., Ebner, M., & Schön, S. (2023). The Analysis of Learning Management System towards Students' Cognitive Learning Outcome: A Systematic Literature Review. *International Journal of Emerging*



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Technologies in Learning (iJET), 18(23), 4–26.

<https://doi.org/10.3991/ijet.v18i23.36443>

Azahra, W., & Muliana, M. P. (2025). Technology's role in lesson plan design for English as a foreign language (EFL) in blended learning. *Wiralodra English Journal*, 9(2), 60–74. <https://doi.org/10.31943/wej.v9i2.452>

Aziz, A., Widiyanto, F., & Purwanto, A. (2024). Analisis Penggunaan Learning Management System sebagai media Pembelajaran pada Mahasiswa Tahun pertama. *Jurnal Studi Guru Dan Pembelajaran*, 7(1), 13–27. <https://doi.org/10.30605/jsqp.7.1.2024.3354>

Balaban, I., & Sobodić, A. (2022). From Teachers' Perspective: Can an Online Digital Competence Certification System Be Successfully Implemented in Schools? *Applied Sciences*, 12(8), 3785. <https://doi.org/10.3390/app12083785>

Barrot, J. S. (2021). K to 12 curriculum reform in the Philippines: Towards making students future ready. *Asia Pacific Journal of Education*, 43(4), 1193–1207. <https://doi.org/10.1080/02188791.2021.1973959>

Belloula, S. (2025). Empowering educators: Leveraging AI to revolutionize lesson planning. *International Journal of Research in Education and Science*, 11(2), 264–280. <https://doi.org/10.46328/ijres.1295>

Bhatt, C., Singh, S., Chauhan, R., Singh, T., & Uniyal, A. (2023). Artificial Intelligence in Current Education: Roles, Applications & Challenges. 2023



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

3rd International Conference on Pervasive Computing and Social Networking (ICPCSN), 241–244.

<https://ieeexplore.ieee.org/document/10265900/>

Bora, B. Y., & Kölemen, C. Ş. (2025). Integrating AI into instructional design: A case study on digital photography education in higher education. *Contemporary Educational Technology*, 17(3), ep583.

<https://doi.org/10.30935/cedtech/16433>

Cadiz, M. C. D., Manuel, L. A. F., Reyes, M. M., Natividad, L. R., & Ibarra, F. P. (2024). Technology integration in Philippine higher education: A content-based bibliometric analysis. *Jurnal Ilmiah Ilmu Terapan Universitas Jambi*, 8(1), 35–47. <https://doi.org/10.22437/jiituj.v8i1.31807>

Campado, R. J., Toquero, C. M. D., & Ulanday, D. M. (2023). Integration of assistive technology in teaching learners with special educational needs and disabilities in the Philippines. *International Journal of Professional Development Learners and Learning*, 5(1), ep2308.

<https://doi.org/10.30935/ijpdll/13062>

Cañete, N. L. (2026). Teachers' technological difficulties and mitigation. *GEO Academic Journal*, 7(2).

<https://doi.org/10.56738/issn29603986.geo2026.7.149>

Carson, J., Benfield, H., & Anderson, M. (2026). Embedding AI in Lesson Planning: Evidence from a Multi-School Research Project. *Proceedings of the*



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

International Conference on Networked Learning, 15.

<https://doi.org/10.54337/nlc.v15.10863>

Chen, J., Zhou, X., Yao, J., & Tang, S. (2025). Application of machine learning in higher education to predict students' performance, learning engagement and self-efficacy: A systematic literature review. *Asian Education and Development Studies, 14*(2), 205–240. <https://doi.org/10.1108/aeds-08-2024-0166>

Chhimwal, G. (2025). Harnessing Machine Learning for Educational Insights: A review and comparative analysis of student performance prediction models. *Journal of Information Systems Engineering & Management, 10*(13s), 643–660. <https://doi.org/10.52783/jisem.v10i13s.2142>

Cornista, M. F. (2025). Voices of the Teachers: Challenges on the immediate removal of administrative tasks of public school teachers to small schools in Marilog District B, Davao City. *International Journal for Multidisciplinary Research, 7*(3). <https://doi.org/10.36948/ijfmr.2025.v07i03.49133>

Crompton, H., & Burke, D. (2023). Artificial intelligence in higher education: The state of the field. *International Journal of Educational Technology in Higher Education, 20*(1). <https://doi.org/10.1186/s41239-023-00392-8>

De Gracia, M. M. B., Cinches, J. C. C., Galang, A. C., & Villanueva, C. J. (2022b). An Analysis of the Perception of Students and Teachers towards the Home-Based Online Learning Modality and the Students' Academic Achievement



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

of a Private School in the National Capital Region (NCR) of the Philippines.

Proceedings of the 13th International Conference on E-Education, E-Business, E-Management and E-Learning (IC4E 2022), 216–223.

<https://doi.org/10.1145/3514262.3514275>

Deckker, D., & Sumanasekara, S. (2025). The role of artificial intelligence in education: Transforming learning and teaching. *EPRA International Journal of Research and Development (IJRD)*, 10(3).

<https://doi.org/10.36713/epra20429>

Dennison, D. V., Ahtisham, B., Chourasia, K., Arora, N., Singh, R., Kizilcec, R. F., Nambi, A. U., Ganu, T., & Vashistha, A. (2025). *Shiksha Copilot: Teacher-AI Collaboration for Curating and Customizing Lesson Plans in Low-Resource Schools*. <https://arxiv.org/pdf/2507.00456>

Diquito, T. J. (2024). Basic Education Curriculum under the Newly Implemented K to 10 (MATATAG) Curriculum in the Philippines: The Case of Science Education. *American Journal of Education and Technology*, 3(3), 123–132.

<https://doi.org/10.54536/ajet.v3i3.3396>

Eden, C. A., Chisom, O. N., & Adeniyi, I. S. (2024). Integrating AI in education: Opportunities, challenges, and ethical considerations. *Magna Scientia Advanced Research and Reviews*, 10(2), 006–013.

<https://doi.org/10.30574/msarr.2024.10.2.0039>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Eka Indah Wahyuni, Fatimah Muthmainnah, Bintang Permana, & Tulas Novalima. (2025). Pengembangan Sistem Informasi Manajemen Sekolah Berbasis Web untuk Meningkatkan Efisiensi Administrasi Pendidikan. *Uranus: Jurnal Ilmiah Teknik Elektro, Sains Dan Informatika*, 3(2), 65–76. <https://doi.org/10.61132/uranus.v3i2.802>
- Enorme, C., Santidad, J., Ledesma, C., Dioneda, M., & Borja, J. (2024). A Systematic Literature Review on ICT Integration: Opportunities and Challenges in Teaching Mathematics. *International Journal on Integrated Education*, 7(4), 151–158. <https://doi.org/10.31149/ijie.v7i4.5327>
- Ersozlu, Z., Taheri, S., & Koch, I. (2024). A review of machine learning methods used for educational data. *Education and Information Technologies*, 29, 22125–22145. <https://doi.org/10.1007/s10639-024-12704-0>
- Esteban, A. J., Calang, K., & Pagador, P. M. E. (2024). A review of practices and digital technology integration in reading instruction and suggestions for the Philippines. *International Journal of Evaluation and Research in Education (IJERE)*, 13(6), 3663. <https://doi.org/10.11591/ijere.v13i6.29856>
- Ettyani. (2025). Transformation of Artificial Intelligent-Based Education: Towards More Personalized and Effective Learning. *Entita: Jurnal Pendidikan Ilmu Pengetahuan Sosial Dan Ilmu-Ilmu Sosial*, 1, 267–278. <https://doi.org/10.19105/ejpis.v1i.19125>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Fabito, B., Trillanes, A., & Sarmiento, J. (2020). Barriers and Challenges of Computing Students in an Online Learning Environment: Insights from One Private University in the Philippines. *International Journal of Computing Sciences Research*, 5(1), 441–458. <https://doi.org/10.25147/ijcsr.2017.001.1.51>
- Fristianingroem, D. A. (2025). Implementation of Learning Management System (LMS)-Based Curriculum in Course and Training Institutions in Tegal City: A Theoretical Study. *Nidhomul Haq: Jurnal Manajemen Pendidikan Islam*, 10(1), 137–150. <https://doi.org/10.31538/ndhq.v10i1.100>
- Frianeza, E. D., Maravilla, H. D., Relox, R. D., Dagaraga, S. J. S. L., Cruz, C. A. D., Solomon, E. H., & Mohammad, N. K. (2024). Challenges in the Philippine Educational System and its Impact Towards Teachers' Instruction Strategies and Professional Growth. *Journal of Pedagogy and Education Science*, 3(01), 63–71. <https://doi.org/10.56741/jpes.v3i01.414>
- Furqon, M., Sinaga, P., Liliyasi, L., & Riza, L. S. (2023). The impact of Learning Management System (LMS) usage on students. *TEM Journal*, 1082–1089. <https://doi.org/10.18421/tem122-54>
- Gamit, A. M. (2023). ICT integration in Elementary School for mathematics subject. *International Journal of Learning Teaching and Educational Research*, 22(2), 432–465. <https://doi.org/10.26803/ijlter.22.2.24>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Haleem, A., Javaid, M., Qadri, M. A., & Suman, R. (2022). Understanding the role of digital technologies in education: A review. *Sustainable Operations and Computers*, 3, 275–285. <https://doi.org/10.1016/j.susoc.2022.05.004>
- Herniawati, A., Holifah, L., & Syakur, A. (2025). The Effectiveness of Learning Management System (LMS) Use in Higher Education. *International Journal Corner of Educational Research*, 4(1), 20–29. <https://doi.org/10.54012/ijcer.v4i1.625>
- Karpouzis, K., Pantazatos, D., Taouki, J., & Meli, K. (2024). Tailoring Education with GenAI: A New Horizon in Lesson Planning. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2403.12071>
- Kasneji, E., Sessler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., Gasser, U., Groh, G., Günnemann, S., Hüllermeier, E., Krusche, S., Kutyniok, G., Michaeli, T., Nerdel, C., Pfeffer, J., Poquet, O., Sailer, M., Schmidt, A., Seidel, T., . . . Kasneji, G. (2023c). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences*, 103, 102274. <https://doi.org/10.1016/j.lindif.2023.102274>
- Kehoe, F. (2023). Leveraging Generative AI Tools for Enhanced Lesson Planning in Initial Teacher Education at Post Primary. *Irish Journal of Technology Enhanced Learning*, 7(2), 172–182. <https://doi.org/10.22554/ijtel.v7i2.124>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Khan, W. (2026). Query-Web AI integrated platform for educational resource management and intelligent answer support. *International Journal for Research in Applied Science and Engineering Technology*, 14(3), 2159–2164. <https://doi.org/10.22214/ijraset.2026.78301>
- Khanzode, P. C. (2026). A Comprehensive review of machine learning Techniques for student Academic Performance Prediction. *International Journal for Research in Applied Science and Engineering Technology*, 14(3), 4223–4228. <https://doi.org/10.22214/ijraset.2026.78840>
- Kovari, A. (2025). A systematic review of AI-powered collaborative learning in higher education: Trends and outcomes from the last decade. *Social Sciences & Humanities Open*, 11, 101335. <https://doi.org/10.1016/j.ssaho.2025.101335>
- Kumar, M. (2025). Role and impact of artificial intelligence in teacher education. *International Journal For Multidisciplinary Research*, 7(3). <https://doi.org/10.36948/ijfmr.2025.v07i03.46908>
- Kusumojati, P. P., & Mediawati, E. (2024). Web-Based Asset Management Information Systems in Higher Education. *International Journal of Business, Law, and Education*, 5(1), 398–411. <https://doi.org/10.56442/ijble.v5i1.382>
- Lacorte, E. E. (2025). Balancing Acts: Teachers' Time Management Strategies to Accomplish Multifaceted Tasks. *EPRA International Journal of Environmental Economics Commerce and Educational Management*, 127–135. <https://doi.org/10.36713/epra23729>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Lagawid, E. L. (2024). Overloading in Elementary School Teachers: A Review. *Cognizance Journal of Multidisciplinary Studies*, 4(11), 289–299. <https://doi.org/10.47760/cognizance.2024.v04i11.016>
- Laksono, C. F., Putri, A. E. P., & Anggraini, R. (2024). Effective Methods for Implementing Artificial Intelligence (AI) in Enhancing Teacher Expertise in Learning. *JATI (Jurnal Mahasiswa Teknik Informatika)*, 9(1), 750–772. <https://doi.org/10.36040/jati.v9i1.12479>
- Lampou, R. (2023). The integration of artificial intelligence in education: Opportunities and challenges. *Review of Artificial Intelligence in Education*, 4(00), e15. <https://doi.org/10.37497/rev.artif.intell.educ.v4i00.15>
- Leong, W. Y., Leong, Y. Z., & Leong, W. S. (2024). Artificial intelligence in education. *IET Conference Proceedings*, 2024(22), 183–184. <https://digital-library.theiet.org/doi/10.1049/icp.2024.4341>
- Li, J.(2025). Enhancing learning through an adaptive web-based educational search framework integrating natural language processing and machine learning techniques. *DiscoverComputing*, 28(1). <https://doi.org/10.1007/s10791-025-09732-w>
- Lucaberte, M. D. G., & Sanchez, R. E. (2024). The Integration of Educational technology in teaching Filipino in Taganito National High School, Claver, Surigao del Norte, Division of Surigao del Norte: A program. *Cognizance*



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Journal of Multidisciplinary Studies, 4(5), 111–116.

<https://doi.org/10.47760/cognizance.2024.v04i05.009>

Mannan, S. S., Latorre, N. J., Kiram, D. a. A., Julaspi, J. M. R., Baddiri, S. P., Jul-Ambri, M. J., & Ahamad, A. A. (2025). Empowering Women in ICT: A case study of female faculty and students in State Universities and Colleges (SUCs) in Jolo, Sulu, Philippines. *International Journal for Multidisciplinary Research*, 7(1). <https://doi.org/10.36948/ijfmr.2025.v07i01.35246>

Marcial, D. E., Palama, J. C., Bucog, F., & Binarao, S. (2024). Teachers' Profile, Motivation, Aptitude, and Satisfaction in a Virtual Upskilling Environment. *Information Technologies and Learning Tools*, 101(3), 164–180. <https://doi.org/10.33407/itlt.v101i3.5491>

Mastul, A.-R. H., de Vera, C. T., & Jayme, C. B. (2023). Understanding the Use of EduTech in Schools in the Philippines: Recommendations for Effectiveness. *Engineering Science Letter*, 2(02), 37–40. <https://doi.org/10.56741/esl.v2i02.307>

Menta, A. J. (2021). Adopting Technology Environment Organization Framework in Virtual Learning through Cloud Computing of Higher Education Institution. *International Journal of Computing Sciences Research*, 6, 822–841. <https://doi.org/10.25147/ijcsr.2017.001.1.75>

Mohammed, A. B., Maqableh, M., Qasim, D., & AlJawazneh, F. (2024). Exploring the factors influencing academic learning performance using online learning



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

systems. *Heliyon*, 10(11), e32584.

<https://doi.org/10.1016/j.heliyon.2024.e32584>

Montebon, D. R. (2024). Perceptions of Select Filipino Teachers on the Most Essential Learning Competencies (MELCs) of the K to 12 Curriculum. *Community and Social Development Journal*, 25(1), 15–27.

<https://doi.org/10.57260/rcmrj.2024.264463>

Moorhouse, B. L., & Yan, L. (2023). Use of Digital Tools by English Language Schoolteachers. *Education Sciences*, 13(3), 226.

<https://doi.org/10.3390/educsci13030226>

Mujorobin, M. (2025, July 9). An analysis of the implementation of Artificial intelligence (AI) in School-Based Learning. *International Journal of Computer and Information System*

(IJCIS) <https://www.ijcis.net/index.php/ijcis/article/view/238/204>

Mujiono, M. (2023). Educational Collaboration: Teachers and Artificial Intelligence.

Jurnal Kependidikan: Jurnal Hasil Penelitian Dan Kajian Kepustakaan Di Bidang Pendidikan, Pengajaran, Dan Pembelajaran, 9(2), 618–632.

<https://doi.org/10.33394/jk.v9i2.7801>

Nagrama, N. D. C., Lingating, M. L. D., Calleno, J. T., & Rato, R. K. A. (2024). Web-based Document Management System. *International Journal of Science and Applied Information Technology*, 13(3), 16–21.

<https://doi.org/10.30534/ijsait/2024/031332024>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Napte, S., & Singh, D. (2025). Artificial Intelligence in Education: A comprehensive systematic review of current trends and future directions. *International Journal on Science and Technology*, 16(2). <https://doi.org/10.71097/ijSAT.v16.i2.6286>
- Nguyen, T. T. K., Nguyen, M. T., & Tran, H. T. (2023). Artificial intelligent based teaching and learning approaches: A comprehensive review. *International Journal of Evaluation and Research in Education*, 12(4), 2387–2396. <https://doi.org/10.11591/ijere.v12i4.26623>
- Ningsih, N. R., Sarah, D., Finorita, E. T., Muhtadin, M., Lestari, D. A., & Suryaman, M. (2026). Utilizing artificial intelligence to support teacher administration and improve education quality. *EDUKASIA: Jurnal Pendidikan dan Pembelajaran*, 7(1), 71–82. <https://doi.org/10.62775/edukasia.v7i1.1763>
- Ningsih, S., Suhardi, M., Sari, I. N., Wulan, D. S., Fadillah, M., & Amalia, N. (2025). Implementasi Learning Management System (LMS) di Lingkungan Pendidikan Tinggi. *EDUTECH: Jurnal Inovasi Pendidikan Berbantuan Teknologi*, 5(2), 422–432. <https://doi.org/10.51878/edutech.v5i2.6370>
- Nuong Deri, M., Singh, A., Zaazie, P., & Anandene, D. (2024). Leveraging Artificial Intelligence in Higher Educational Institutions: A Comprehensive Overview. *Revista De Educación Y Derecho*, (30). <https://doi.org/10.1344/REYD2024.30.45777>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Onesi-Ozigagun, O., Ololade, Y. J., Eyo-Udo, N. L., & Ogundipe, D. O. (2024). Revolutionizing education through AI: A comprehensive review of enhancing learning experiences. *International Journal of Applied Research in Social Sciences*, 6(4), 589–607. <https://doi.org/10.51594/ijarss.v6i4.1011>
- Orena, Z., & Gamutan, L. (2025). Extent of Workload Challenges and Multitasking Skills Among Teachers and their Performance. *Psychology and Education: A Multidisciplinary Journal*, 42(3), 514–517. <https://doi.org/10.70838/pemj.420308>
- Portugues, T. M. (2026). Teachers' information and communication technology (ICT) skill, technology integration and learners' performance. *International Journal on Science and Technology*, 17(1). <https://doi.org/10.71097/ijssat.v17.i1.10066>
- Rafiq-uz-Zaman, M. (2023). The Impact of Digital Literacy on Students' Learning Outcomes: A Comprehensive Review. *Inverge Journal of Social Sciences*, 2(2), 194–205. <https://doi.org/10.63544/ijss.v2i2.210>
- Rahayu, D. D., Suherni, H. P., Al-Ma'mun, M. S., Alkahfi, F. A., Hamid, F., & Faridah. (2025). Indonesian EFL Teachers' perspectives on AI for Lesson Planning. *English Education Jurnal Tadris Bahasa Inggris*, 18(2), 374–388. <https://www.semanticscholar.org/paper/Indonesian-EFL-Teachers'-Perspectives-on-AI-for-Rahayu-Suherni/d3268142c0221c53feb5925e7797dffef5fcc09c>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Ramalingam, S., & Maniam, M. (2024). Artificial intelligence trends in education among school administrators in Malaysia. *Proceedings of the International Conference on Education*, 2(1), 75–81.
<https://doi.org/10.32672/pice.v2i1.1320>
- Rekha, R. A. P. (2024). The Learning Management System (LMS) and Student Learning Effectiveness: A Systematic Literature Review. *International Journal of Research Publication and Reviews*, 5(6), 1857–1861.
<https://doi.org/10.55248/gengpi.5.0624.1460>
- Reyes, M. S., & San Buenaventura-Gaza, J. (2025). Educational Technology in Elementary Teaching: Philosophies and Practices of Selected Teachers. *Journal of Pedagogy and Education Science*, 4(03), 546–561.
<https://doi.org/10.56741/IISTR.jpes.001088>
- Rian Novita. (2025). AI in Lesson Planning: Improving Teacher Efficiency and Instructional Design. *Jurnal Riset Rumpun Ilmu Pendidikan*, 4(2), 192–202.
<https://doi.org/10.55606/jurripen.v4i2.5560>
- Roque, J. P. (2023). Modular distance learning in the area of education during the new normal: A Systematic Review. *AIDE Interdisciplinary Research Journal*, 3, 253–268. <https://doi.org/10.56648/aide-irj.v3i1.67>
- Salama, W. M., & Aly, M. H. (2024). Web-Based Integrated School Management System for Enhanced Student-Teacher and Resource Accessibility. *2024 34th International Conference on Computer Theory and Applications*



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

(ICCTA), 115–122. <https://www.semanticscholar.org/paper/Web-Based-Integrated-School-Management-System-for-Salama-Aly/3fa8018906628d0937bc60b2f026bf625d6e7902>

Sanchez, L., Penarreta, J., & Poma, X. S. (2024). Learning management systems for higher education: A brief comparison. *Discover Education*, 3(1). <https://doi.org/10.1007/s44217-024-00143-5>

Saracostti, M., De Toro, X., Rossi, A., Lara, L., & Sotomayor, M. B. (2022). Implementation of a web-based system to measure, monitor, and promote school engagement strategies: A Chilean experience. *Frontiers in Psychology*, 13, 980902. <https://doi.org/10.3389/fpsyg.2022.980902>

Setyaningsih, E., Asrori, M., Ngadiso, Sumardi, Zainnuri, H., & Hariyanti, Y. (2024). Exploring High School EFL Teachers' Experiences with Magic School AI in Lesson Planning: Benefits and Insights. *Voices of English Language Education Society*, 8(3). <https://doi.org/10.29408/veles.v8i3.27700>

Sultan Bader Aljehani. (2024). Enhancing Student Learning Outcomes: The Interplay of Technology Integration, Pedagogical Approaches, Learner Engagement, and Leadership Support. *Educational Administration: Theory and Practice*, 30(4), 418–437. <https://doi.org/10.53555/kuey.v30i4.1485>

Systematic Literature Review: Machine Learning in Education to Predict Student Performance. (2022, September 15). *IEEE Conference Publication | IEEE Xplore*. <https://ieeexplore.ieee.org/document/9967900/>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Tanucan, J. C. M., Alejandro, B. A., & Corcino, R. B. (2023). Towards an Enhanced Implementation of Printed Modular Distance Learning in the Philippines: A Meta-Synthesis. *International Journal of Learning Teaching and Educational Research*, 22(3), 341–358. <https://doi.org/10.26803/ijlter.22.3.21>
- Teräs, M. (2022). Education and technology: Key issues and debates. *International Review of Education*, 68(4), 635–636.
<https://doi.org/10.1007/s11159-022-09971-9>
- THE USE OF WEB TOOLS IN EDUCATION: A CASE STUDY IN A SECONDARY SCHOOL IN MONTENEGRO. (2025). *International Journal of Management Trends: Key Concepts and Research*, 4(2), 95–106.
<https://doi.org/10.58898/ijmt.v4i2.95-106>
- Timotheou, S., Miliou, O., Dimitriadis, Y., Sobrino, S. V., Giannoutsou, N., Cachia, R., Monés, A. M., & Ioannou, A. (2022). Impacts of digital technologies on education and factors influencing schools' digital capacity and transformation: A literature review. *Education and Information Technologies*, 28(6), 6695–6726. <https://doi.org/10.1007/s10639-022-11431-8>
- Tjiptady, B. C., Rohman, M., Rahman, R. Z., Elmunsyah, H., & Tarecha, R. I. (2024). Exploring Masterweb's Effectiveness in Optimizing Digital Based Learning and Education Management. *Ultima InfoSys: Jurnal Ilmu Sistem Informasi*, 15(2), 74–79. <https://doi.org/10.31937/si.v15i2.3596>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Tolentino, J. C. G., Campo, J. C., Licup, T. O., Lumba, E. G., & Lingat, E. G. (2024). Technology Integration in Physical Education Curriculum Implementation in the Philippines: A Scoping Review. *International Journal of Multidisciplinary: Applied Business and Education Research*, 5(7), 2906–2918. <https://doi.org/10.11594/ijmaber.05.07.37>
- Tongka, F. A., Novian, C., Putra, H. I., Utami, E., & Fattah, H. A. (2025). Applying artificial intelligence technology to web-based school management system administrative task automation. *International Journal of Applied Research and Sustainable Sciences*, 3(5), 413–430. <https://doi.org/10.59890/ijarss.v3i5.37>
- Ullrich, A., Vladova, G., Eigelshoven, F., & Renz, A. (2022). Data mining of scientific research on artificial intelligence in teaching and administration in higher education institutions: A bibliometrics analysis and recommendation for future research. *Discover Artificial Intelligence*, 2, 1–18. <https://doi.org/10.1007/s44163-022-00031-7>
- UNESCO. (2023). *Technology in education: A tool on whose terms? Global Education Monitoring Report 2023*. Paris, France: UNESCO. <https://www.unesco.org/gem-report/en/publication/technology-education>
- Utami, L., Andri Majid, Bobur Sobirov, & Jaffer Basha. (2024). Optimizing Education Through Website-Based Learning Media: Implementation in Junior High Schools. *Ceddi Journal of Education*, 3(2), 16–27. <https://doi.org/10.56134/cje.v3i2.104>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Valdez, P. N., Lapinid, M. R., & Chua, V. C. (2024). Agency at Work: A Metaphor Analysis of Lesson Planning Among English Language Teachers in the Philippines. *TEFLIN Journal*, 35(2), 352–369. <https://doi.org/10.15639/teflinjournal.v35i2/352-369>
- Valverde-Berrocso, J., Acevedo-Borrega, J., & Cerezo-Pizarro, M. (2022). Educational Technology and Student Performance: A Systematic Review. *Frontiers in Education*, 7. <https://doi.org/10.3389/feduc.2022.916502>
- Villaseñor, R. A. M. (2025). Challenges and Dilemmas of Digitalization in Philippine Education: A Grassroots Perspective. *Journal of Public Administration and Governance*, 14(2), 232. <https://doi.org/10.5296/jpaq.v14i2.22325>
- Win Min Zaw, & Su Su Hlaing. (2024). Bridging the Educational Gap: The Role of Digital Learning Platforms in Developing Countries. *International Journal of Educational Development*, 1(1), 11–15. <https://doi.org/10.61132/ijed.v1i1.122>
- Wu, X. (2023). Exploring the effects of digital technology on deep learning: A meta-analysis. *Education and Information Technologies*, 29(1), 425–458. <https://doi.org/10.1007/s10639-023-12307-1>
- Yin, S. S. Y., & Mohamad, M. (2023). Unleashing the Potential: A Systematic Review of Teachers' Perspectives on Enhancing Teaching Practices through Digital Tools. *International Journal of Academic Research in Business and Social Sciences*, 13(8). <https://doi.org/10.6007/ijarbss/v13-i8/18202>



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

- Yu, Z. (2022). Sustaining Student Roles, Digital Literacy, Learning Achievements, and Motivation in Online Learning Environments during the COVID-19 Pandemic. *Sustainability*, 14(8), 4388. <https://doi.org/10.3390/su14084388>
- Yulianti, D., Suriansyah, A., Harsono, A. M. B., Ridhaningtyas, L. P., & Noorhapizah, N. (2025). Implementation of Web-Based Teaching Media as an Effort to Utilize Digital Technology in Learning. *AMPLITUDO Journal of Science & Technology Innovation*, 4(1), 20–25. <https://doi.org/10.56566/amplitudo.v4i1.301>
- Zalun, J. G. (2023). The Teachers' Utilization of the Most Essential Learning Competencies (MELCS) and its Relation to the Learning Development of Grade Six Pupils in a Public School in the Philippines: Basis for a Proposed Program. *International Journal of Multidisciplinary: Applied Business and Education Research*, 4(6), 1888–1903. <https://doi.org/10.11594/ijmaber.04.06.15>
- Zhao Ma, X., Ertmer, P. A., Pelgrumen, C. P. M., Watsonta, J. R., & Sengha Tanu, M. C. (2024). The Impact of Technology Integration on Student Learning Outcomes. *Journal of Teaching and Learning*, 1(1), 73–90. <https://doi.org/10.71305/jtl.v1i1.108>